

✓ Matplotlib is a low level graph plotting library in python that serves as a visualization utility.

Matplotlib is open source and we can use it freely.

The source code for Matplotlib is located at this github repository

<https://github.com/matplotlib/matplotlib>

1. Install Matplotlib using pip

```
pip install matplotlib
```

```
➡ Requirement already satisfied: matplotlib in /usr/local/lib/python3.11/dist-packages (3.10.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.11/dist-packages (from matplo
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.11/dist-packages (from matplotlib
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.11/dist-packages (from matpl
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.11/dist-packages (from matpl
Requirement already satisfied: numpy>=1.23 in /usr/local/lib/python3.11/dist-packages (from matplotlib)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.11/dist-packages (from matplot
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.11/dist-packages (from matplotlib) (
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.11/dist-packages (from matplo
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.11/dist-packages (from ma
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages (from python-dateuti
```

2. Use Matplotlib through import

```
import matplotlib
```

3. Import Sub Module pyplot of Matplotlib because most of the visualization tools available in this sub module.

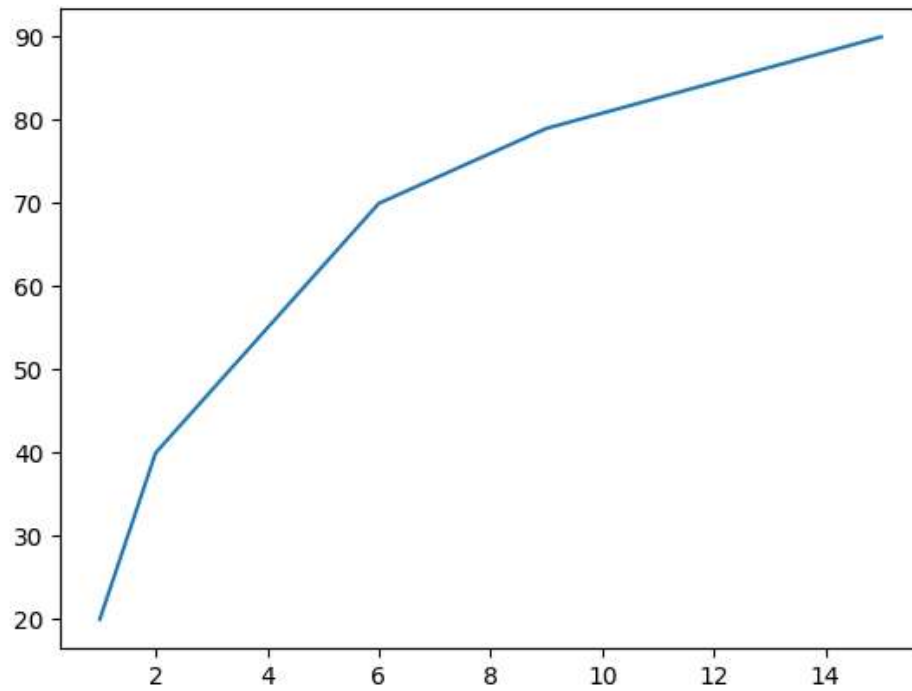
```
import matplotlib.pyplot as plt
```

```
from matplotlib import pyplot as plt
```

4. Here, we create data using list

```
x = [1,2,6,9,15]
y = [20,40,70,79,90]
plt.plot(x,y)
```

↔ [matplotlib.lines.Line2D at 0x7eb9967cd9d0>]



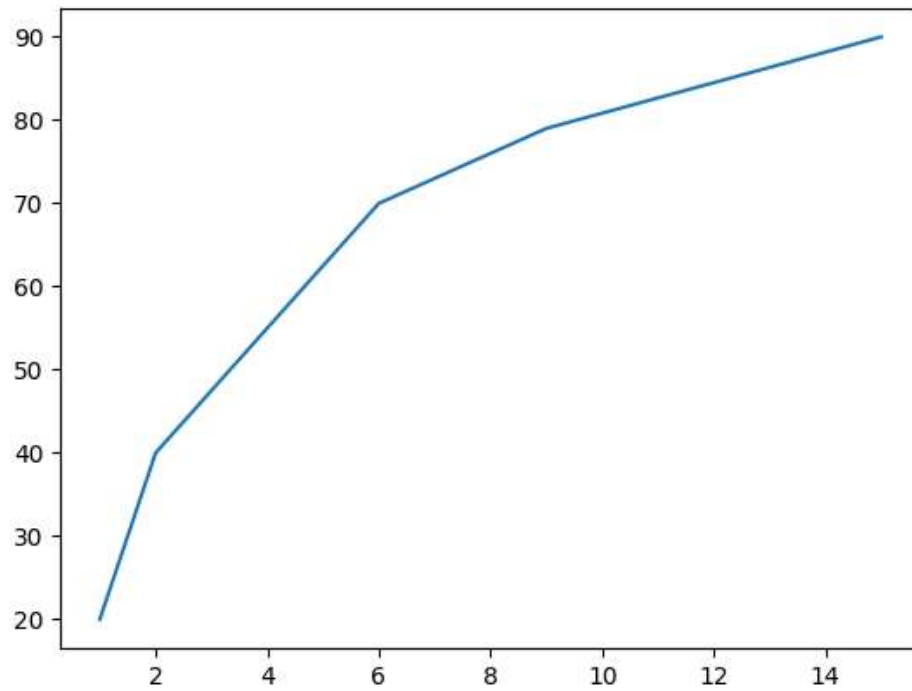
5. Import numpy to create data in the form of numeric array and perform visualization

```
import numpy as np
```

6. Here, we create data using numpy array uses list

```
x = [1,2,6,9,15]
y = [20,40,70,79,90]
x = np.array(x)
y = np.array(y)
plt.plot(x,y)
```

➡ [matplotlib.lines.Line2D at 0x7eb994532390>]

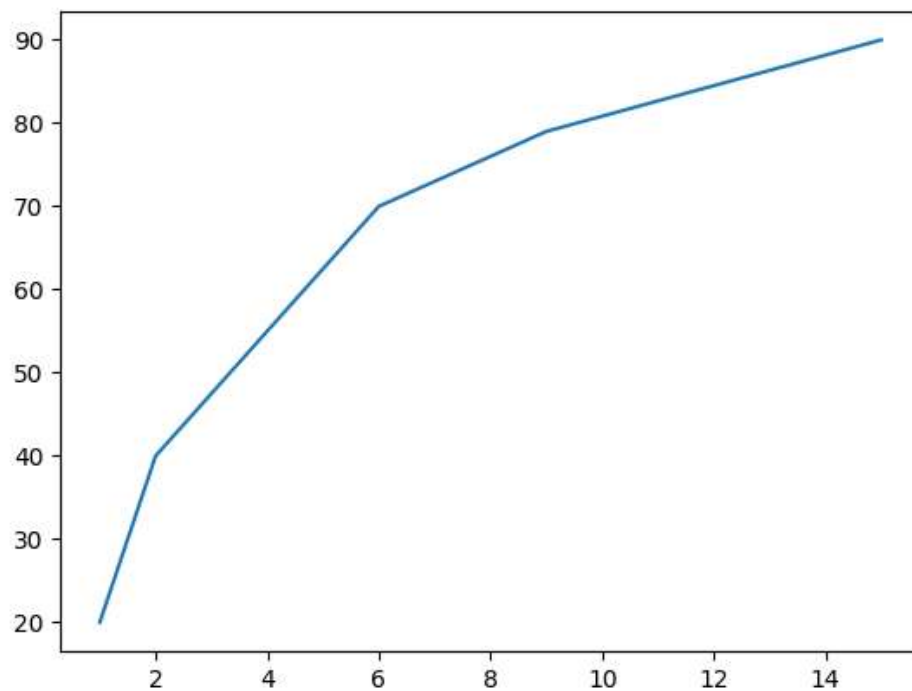


6. Here, we create data using tuple

7. Here, we create data using tuple

```
x = (1,2,6,9,15)
y = (20,40,70,79,90)
plt.plot(x,y)
```

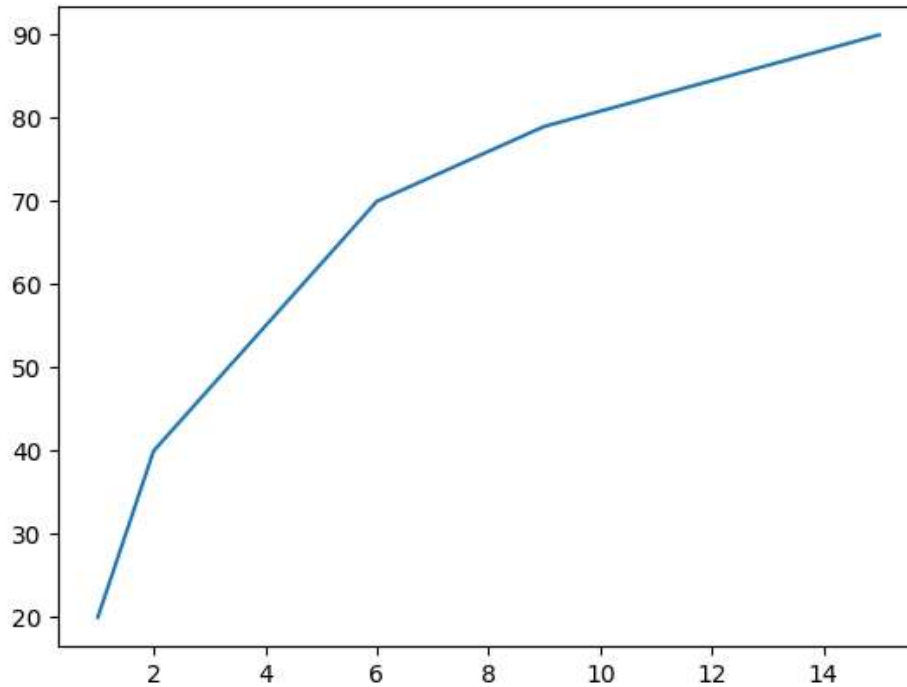
➡ [matplotlib.lines.Line2D at 0x7eb97cd6c0d0>]



8. Numpy array contain tuple and perform visualization

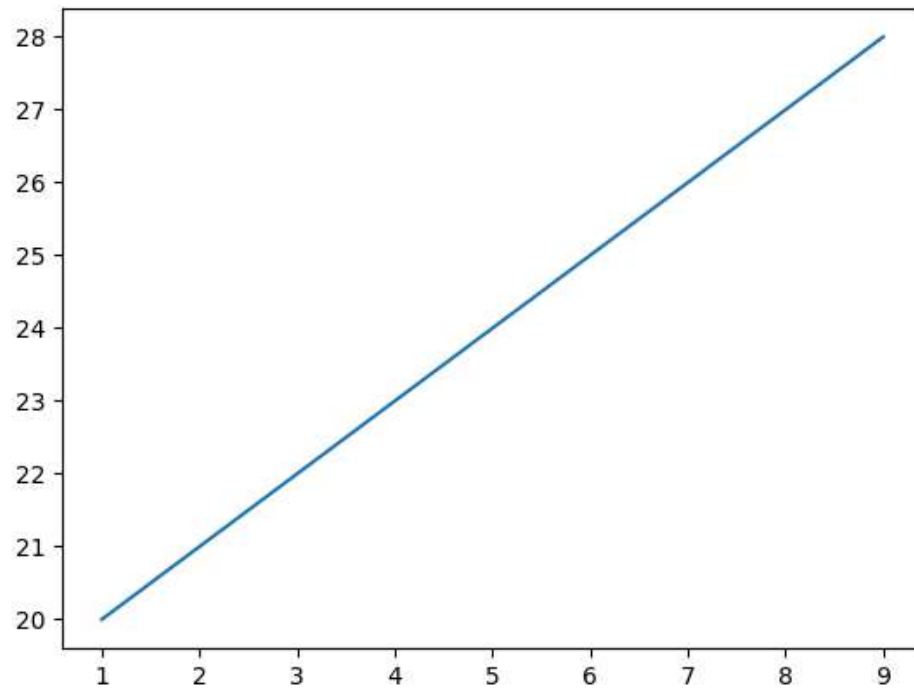
```
x = (1,2,6,9,15)
y = (20,40,70,79,90)
x1=np.array(x)
y1=np.array(y)
plt.plot(x1,y1)
```

→ [[matplotlib.lines.Line2D](#) at 0x7eb97cd43c10>]



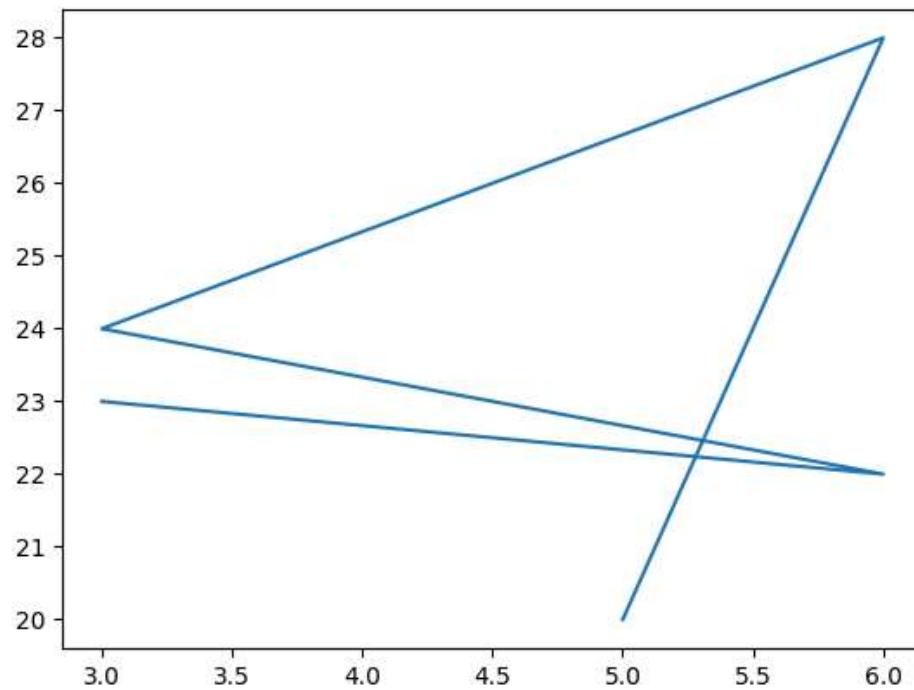
```
#using range function in numpy to create data
x = np.arange(1,10,2)
y = np.arange(20,30,2)
plt.plot(x,y)
```

➡ [matplotlib.lines.Line2D at 0x7eb99441a7d0>]



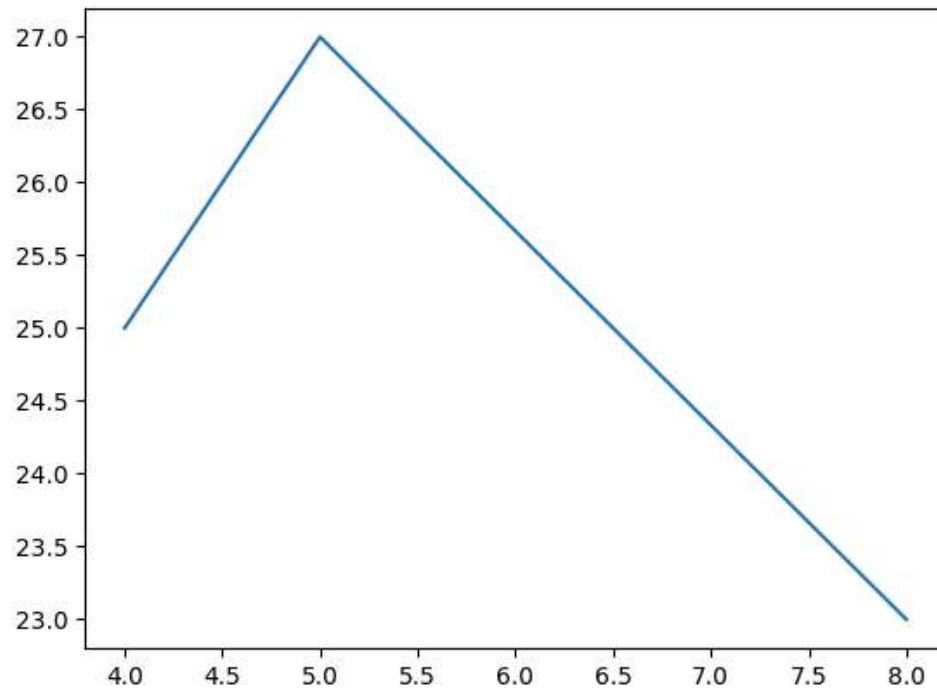
```
#using random module in numpy to create data
x = np.random.randint(1,10,5)
y = np.random.randint(20,30,5)
plt.plot(x,y)
```

➡ [matplotlib.lines.Line2D at 0x7eb994484810>]



```
#using random module in numpy to create data
x = np.random.randint(1,10,3)
y = np.random.randint(20,30,3)
plt.plot(x,y)
```

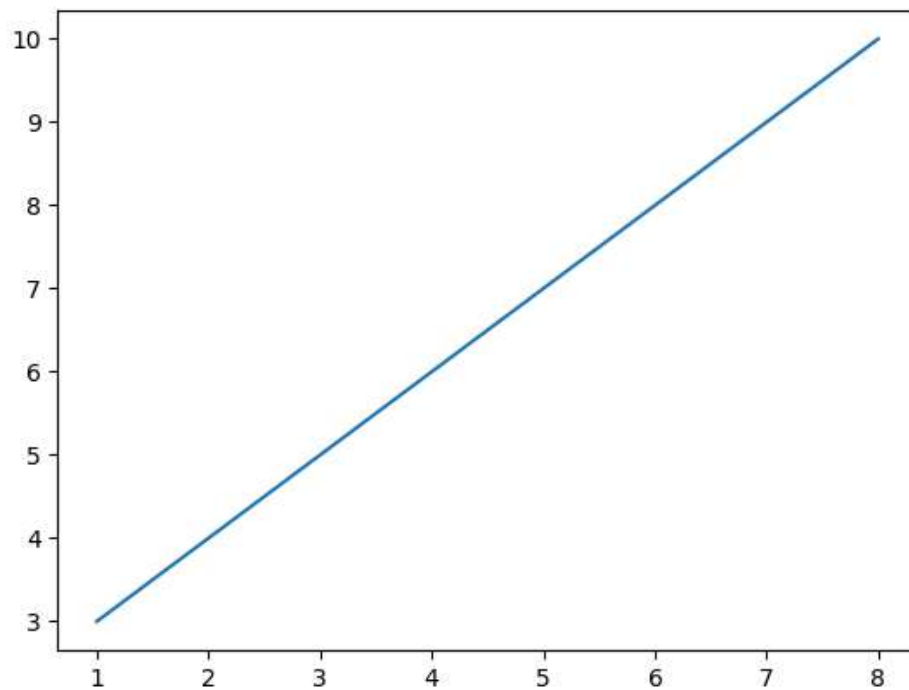
↔ [matplotlib.lines.Line2D at 0x7eb9940f2090>]



```
xpoints = np.array([1, 8])  
ypoints = np.array([3, 10])
```

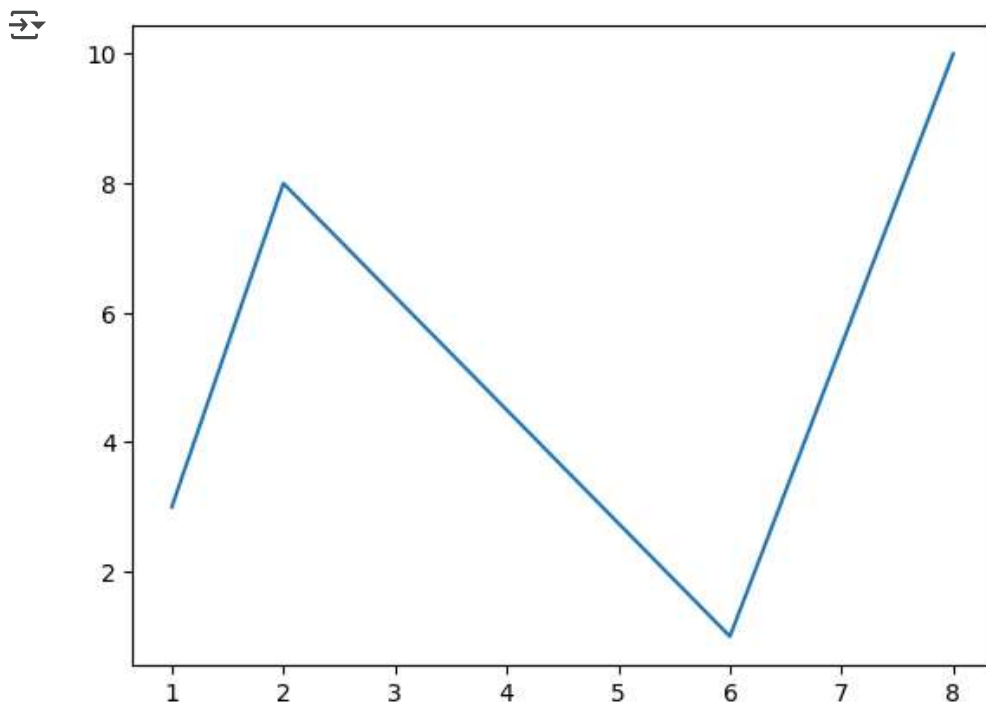
```
plt.plot(xpoints, ypoints)  
plt.show()
```

↔



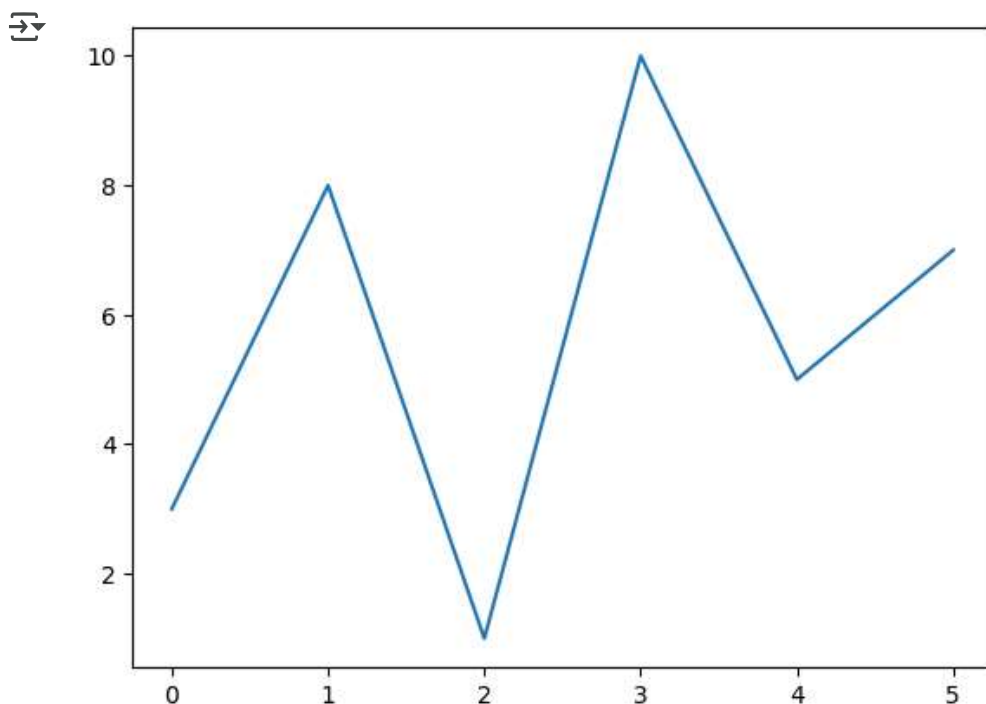
```
xpoints = np.array([1, 2, 6, 8])  
ypoints = np.array([3, 4, 8, 10])
```

```
plt.plot(xpoints, ypoints)  
plt.show()
```



- ✓ If we do not specify the points on the x-axis, they will get the default values 0, 1, 2, 3 etc., depending on the length of the y-points.

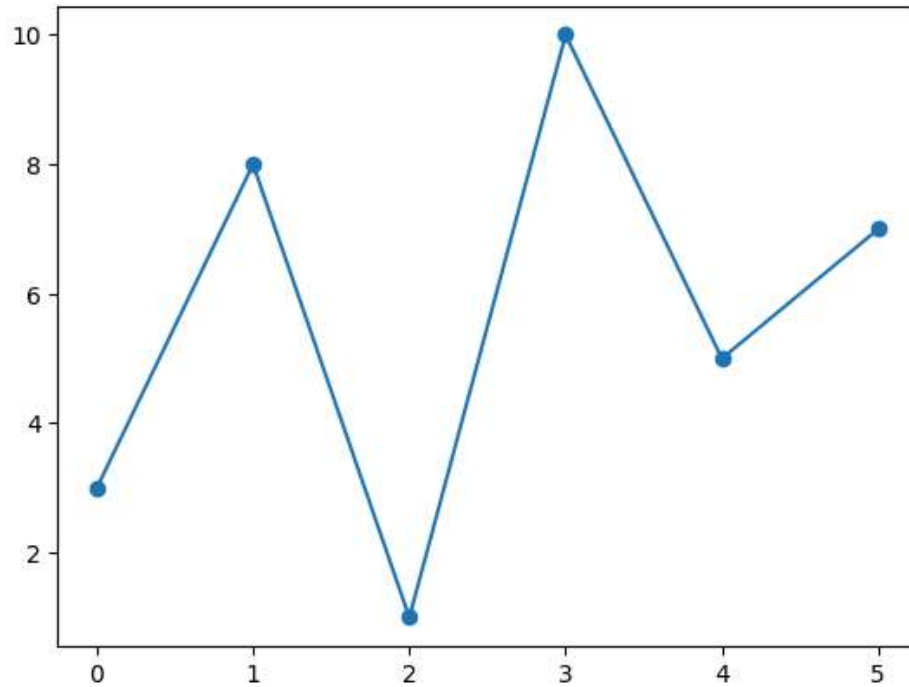
```
ypoints = np.array([3, 8, 1, 10, 5, 7])  
plt.plot(ypoints)  
plt.show()
```



✓ Types of Markers Used

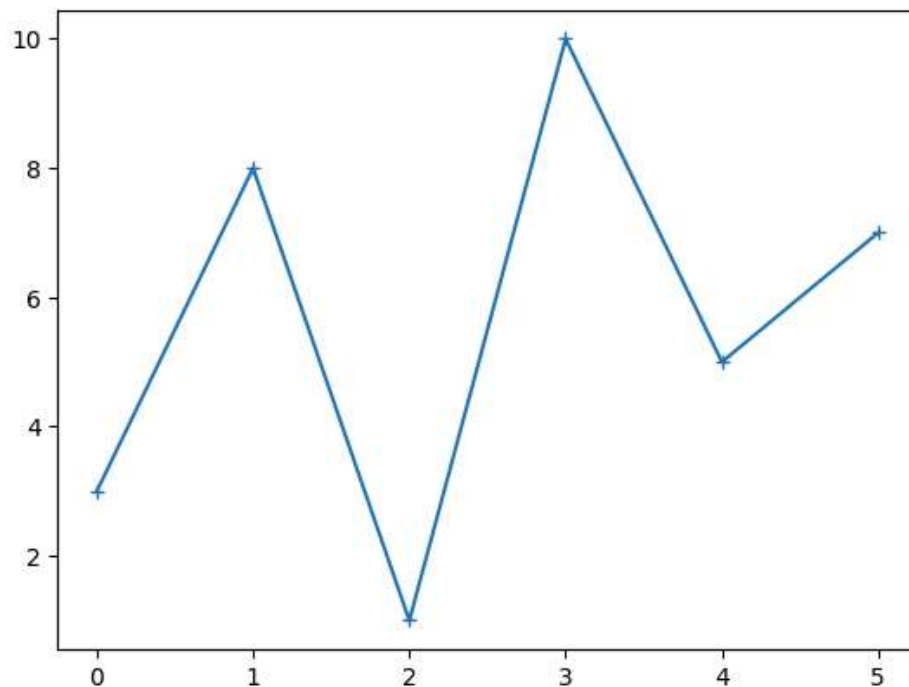
```
ypoints = np.array([3, 8, 1, 10, 5, 7])
```

```
plt.plot(ypoints,marker='o')  
plt.show()
```



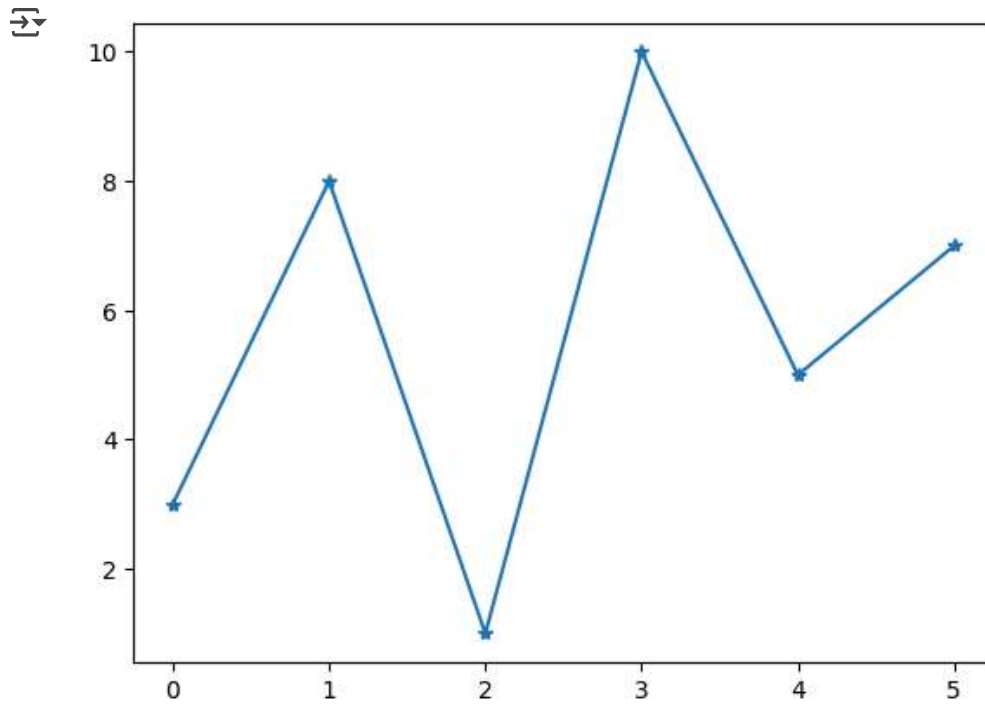
```
ypoints = np.array([3, 8, 1, 10, 5, 7])
```

```
plt.plot(ypoints,marker='+')  
plt.show()
```



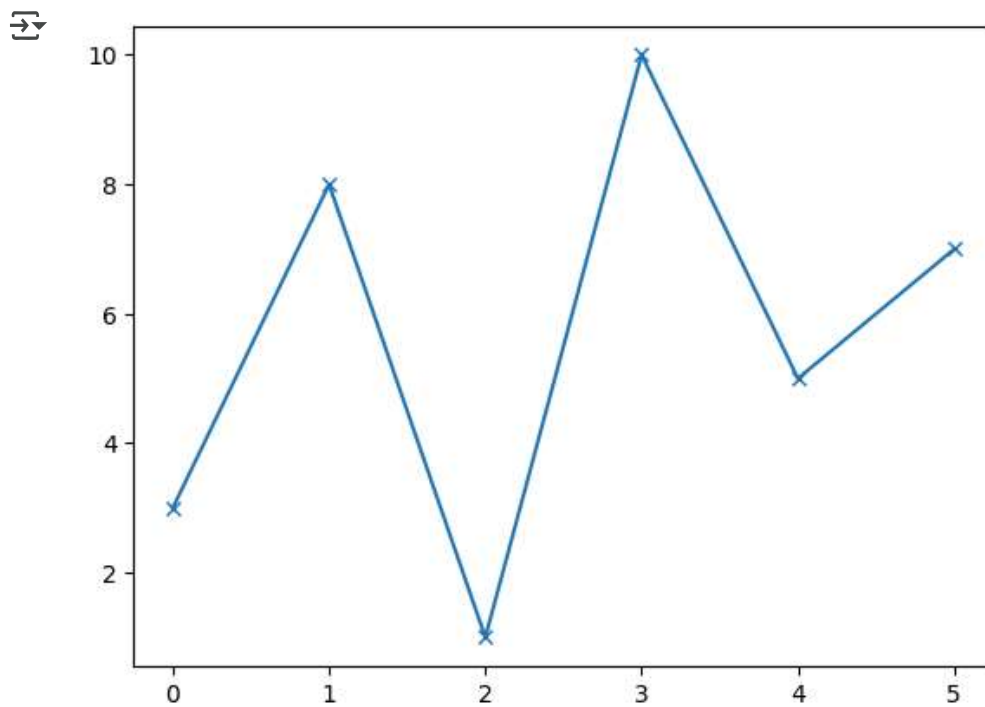

```
ypoints = np.array([3, 8, 1, 10, 5, 7])
```

```
plt.plot(ypoints,marker='*')  
plt.show()
```



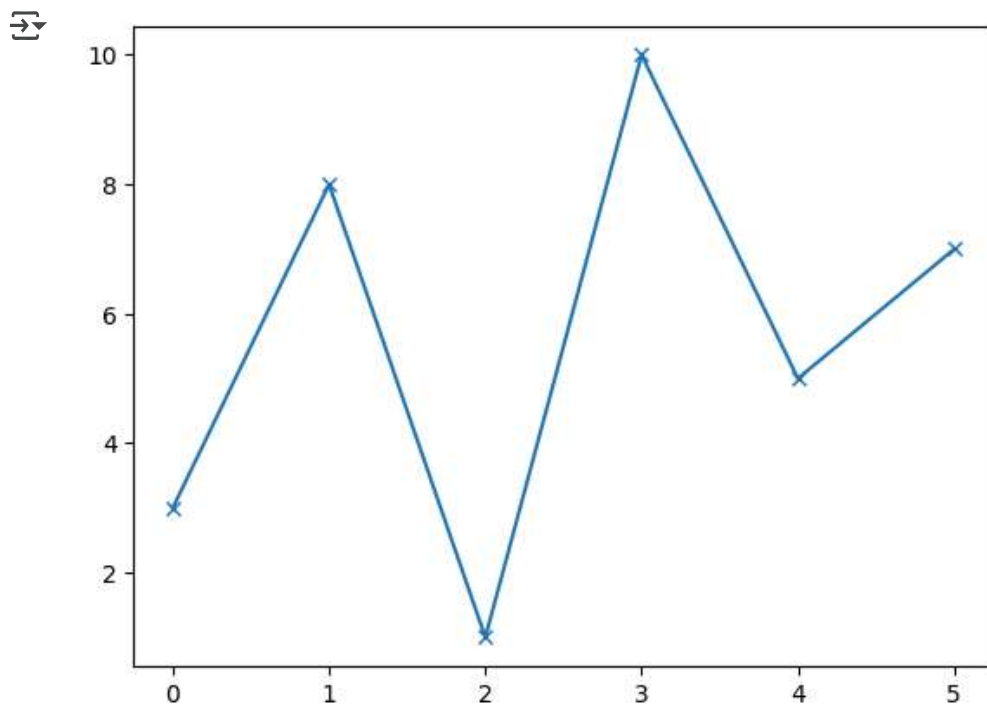
```
ypoints = np.array([3, 8, 1, 10, 5, 7])
```

```
plt.plot(ypoints,marker='x')  
plt.show()
```



```
ypoints = np.array([3, 8, 1, 10, 5, 7])
```

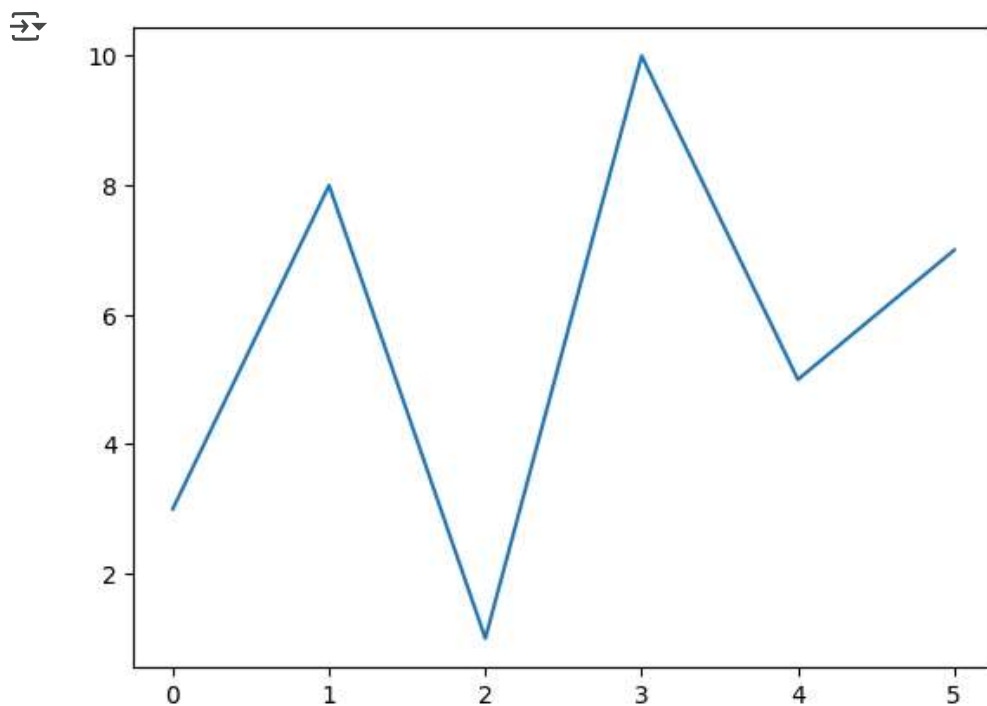
```
plt.plot(ypoints,marker='x')  
plt.show()
```



✓ Line Style

```
ypoints = np.array([3, 8, 1, 10, 5, 7])
```

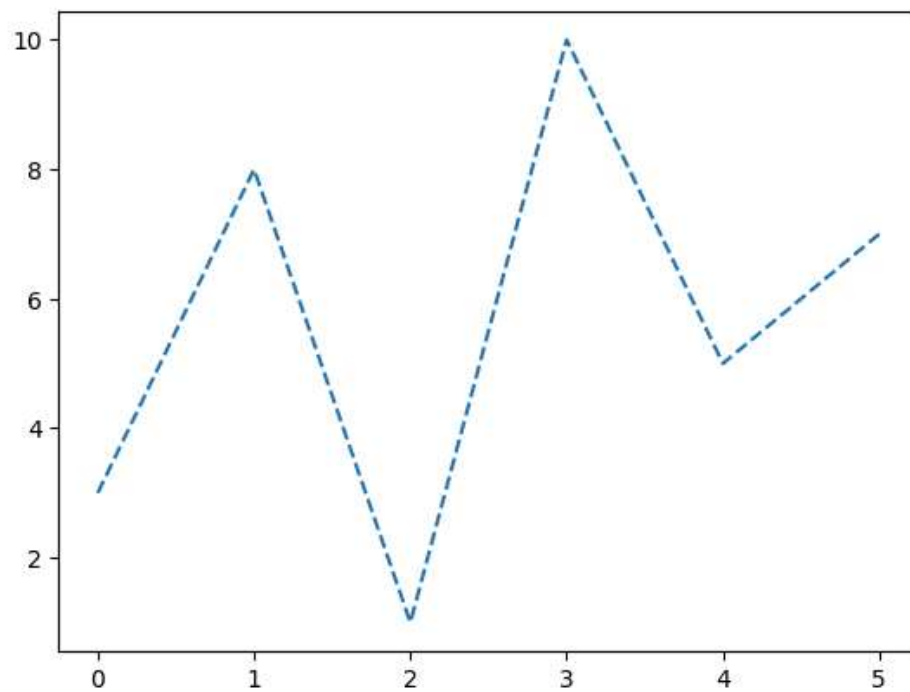
```
plt.plot(ypoints,linestyle='-')  
plt.show()
```



```
ypoints = np.array([3, 8, 1, 10, 5, 7])
```

```
plt.plot(ypoints,linestyle='--')
```

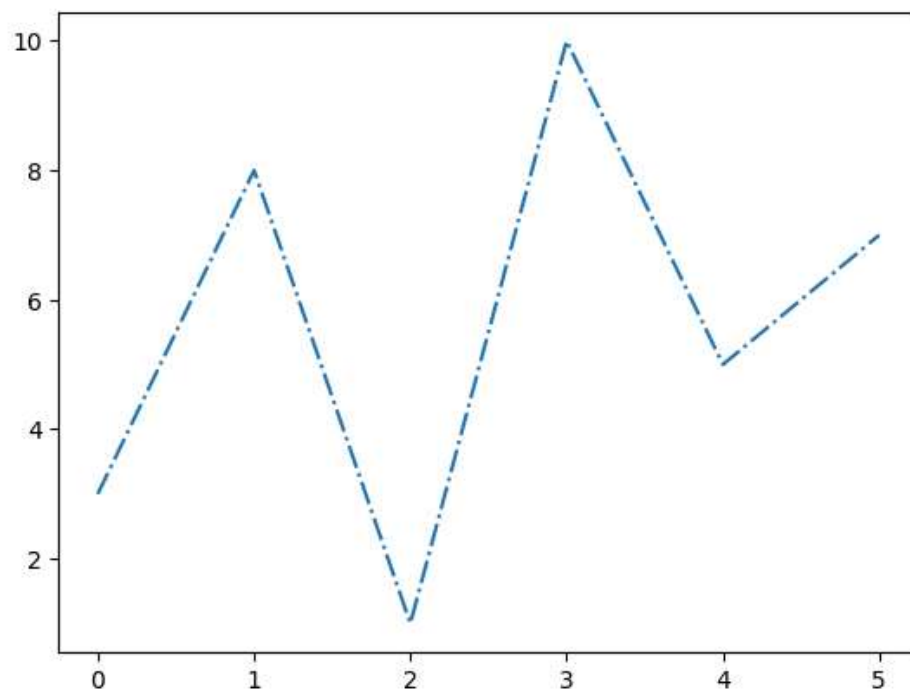
```
plt.show()
```



```
ypoints = np.array([3, 8, 1, 10, 5, 7])
```

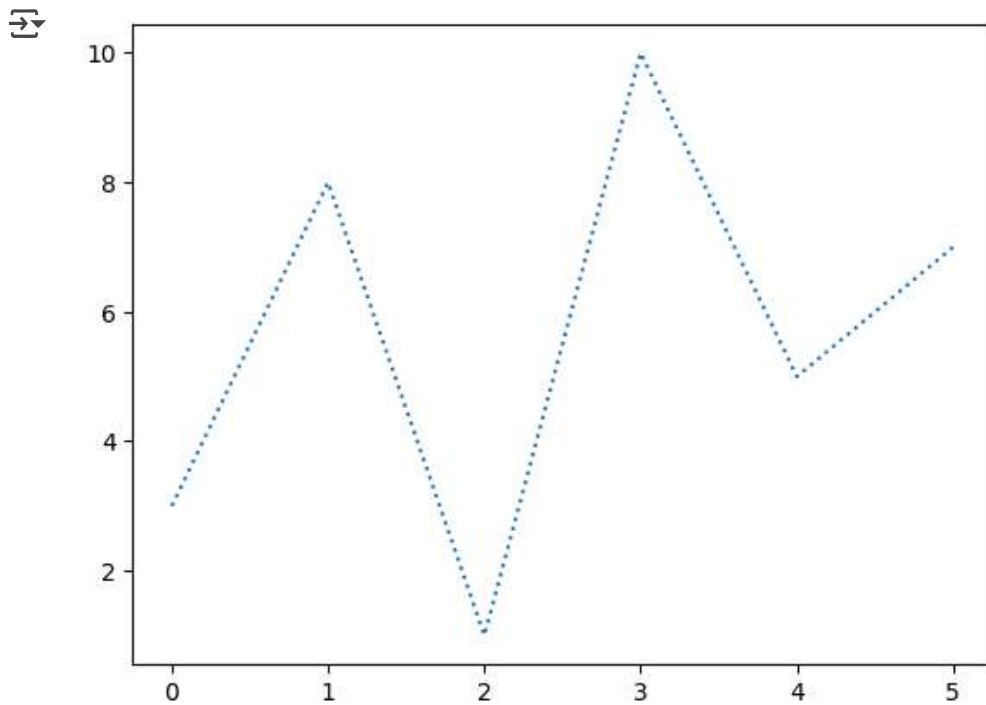
```
plt.plot(ypoints,linestyle='-.')
```

```
plt.show()
```



```
ypoints = np.array([3, 8, 1, 10, 5, 7])
```

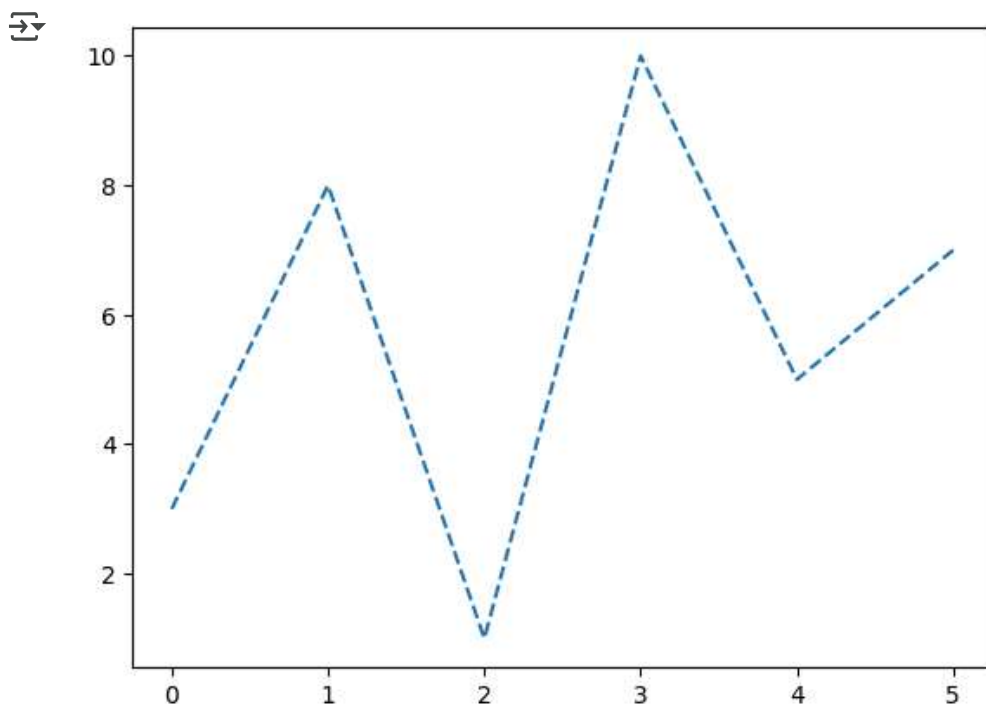
```
plt.plot(ypoints,linestyle=':')  
plt.show()
```



✓ We can use linestyle or ls for type of line used

```
ypoints = np.array([3, 8, 1, 10, 5, 7])
```

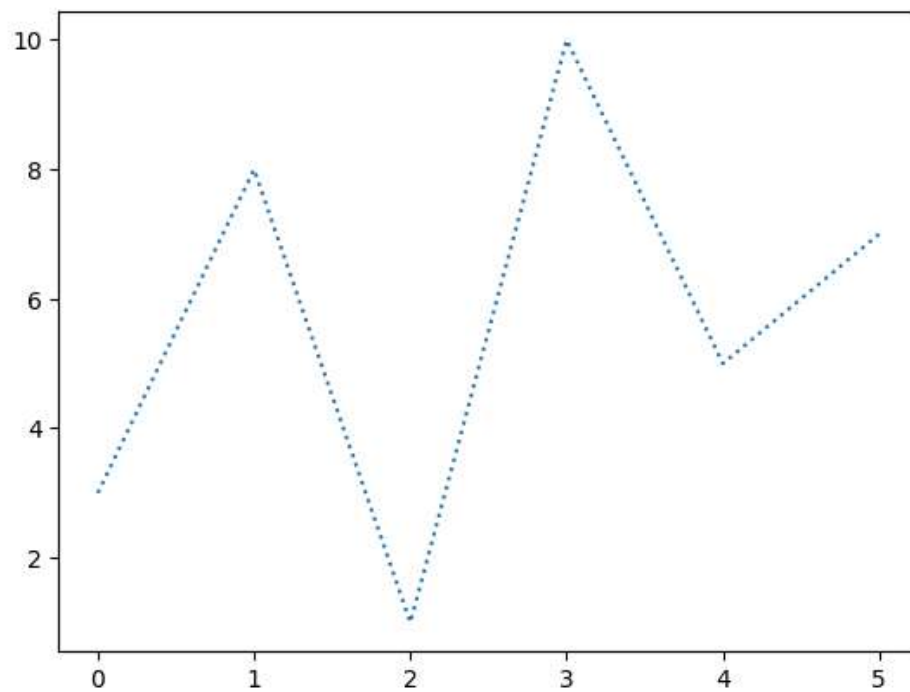
```
plt.plot(ypoints,ls='--')  
plt.show()
```



```
ypoints = np.array([3, 8, 1, 10, 5, 7])
```

```
plt.plot(ypoints,ls='dotted')
```

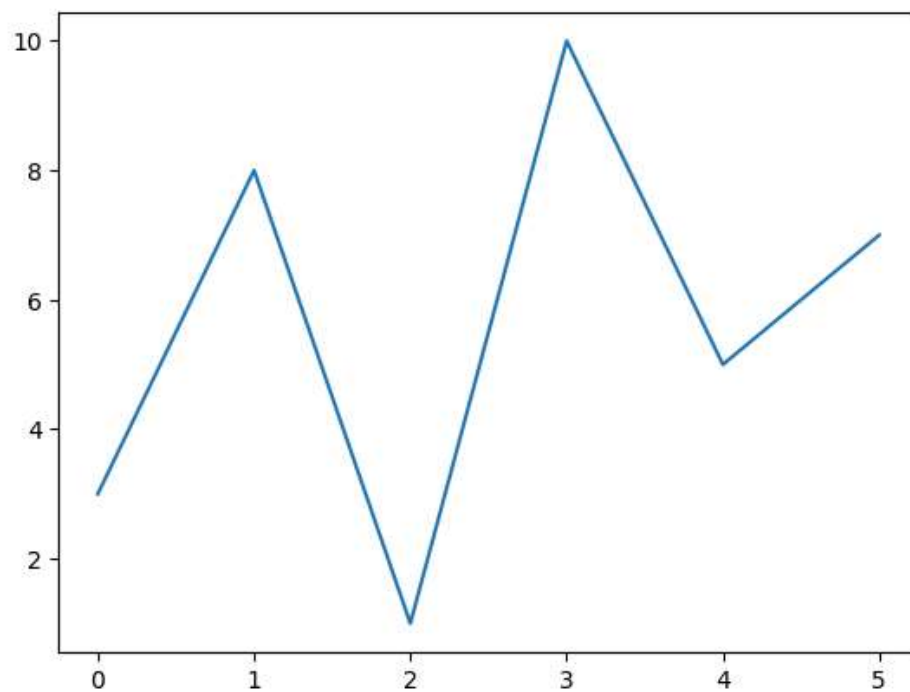
```
plt.show()
```



```
ypoints = np.array([3, 8, 1, 10, 5, 7])
```

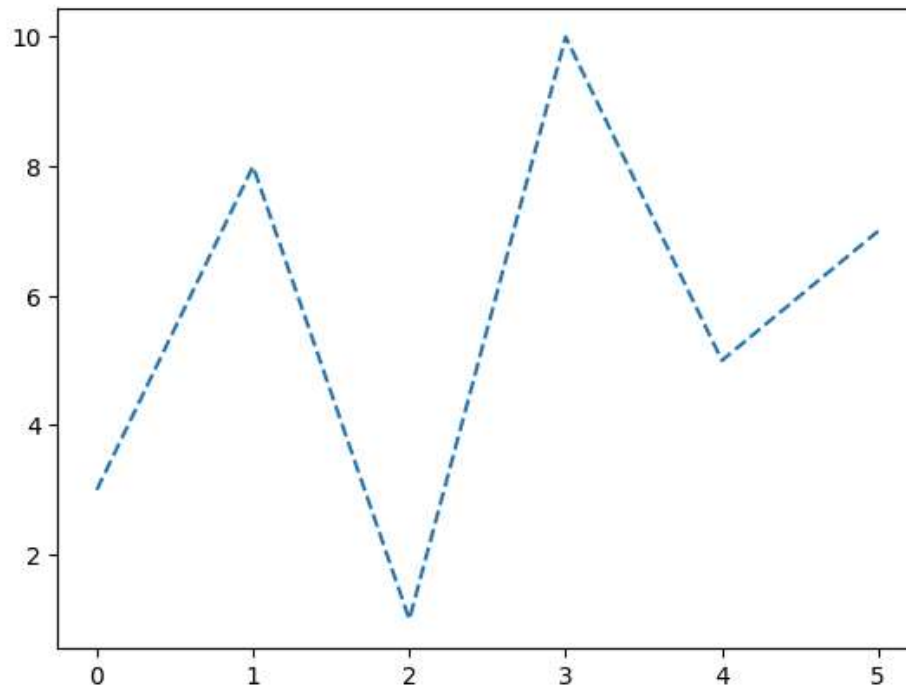
```
plt.plot(ypoints,ls='solid')
```

```
plt.show()
```



```
ypoints = np.array([3, 8, 1, 10, 5, 7])
```

```
plt.plot(ypoints,ls='dashed')  
plt.show()
```

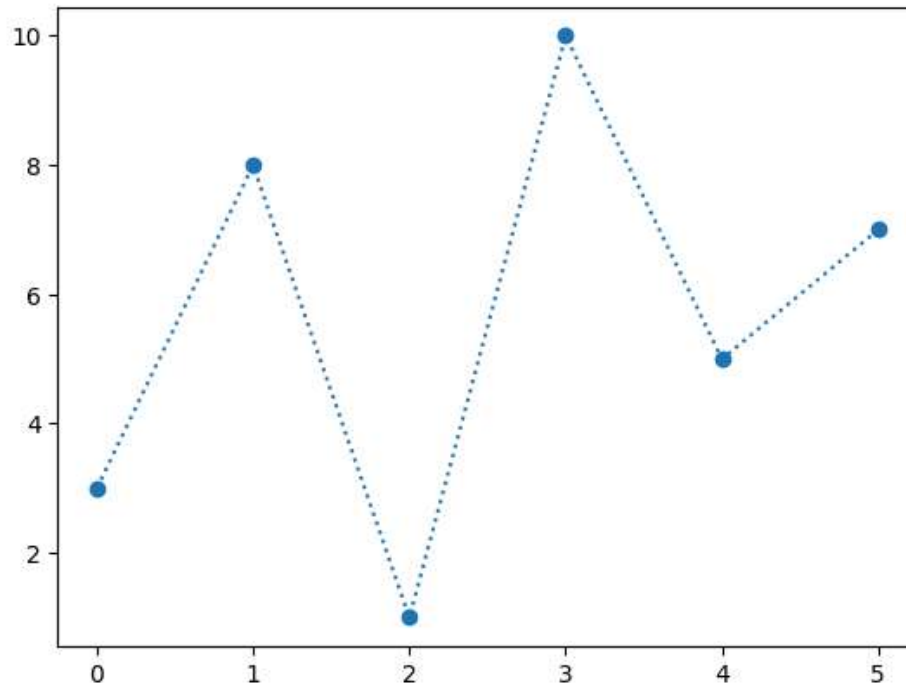


✓ Line Style With Marker

#Here o for marker and : for line style

```
ypoints = np.array([3, 8, 1, 10, 5, 7])
```

```
plt.plot(ypoints,'o:')  
plt.show()
```

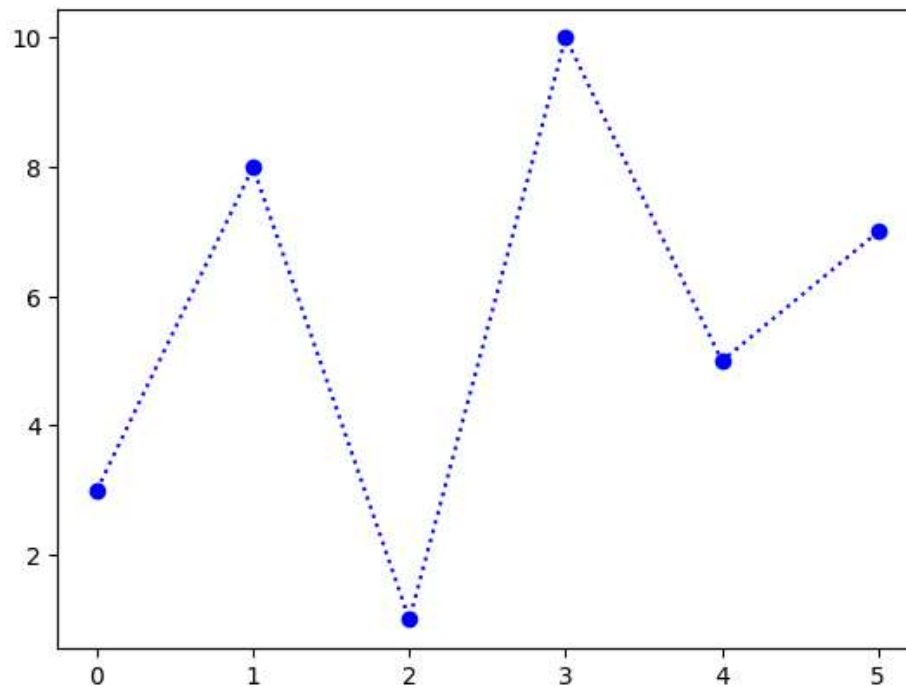


✓ Line Style With Marker And Color of Line and Marker

#Here o for marker and : for line style and b means blue : marker|line|color

```
ypoints = np.array([3, 8, 1, 10, 5, 7])
```

```
plt.plot(ypoints,'o:b') #  
plt.show()
```

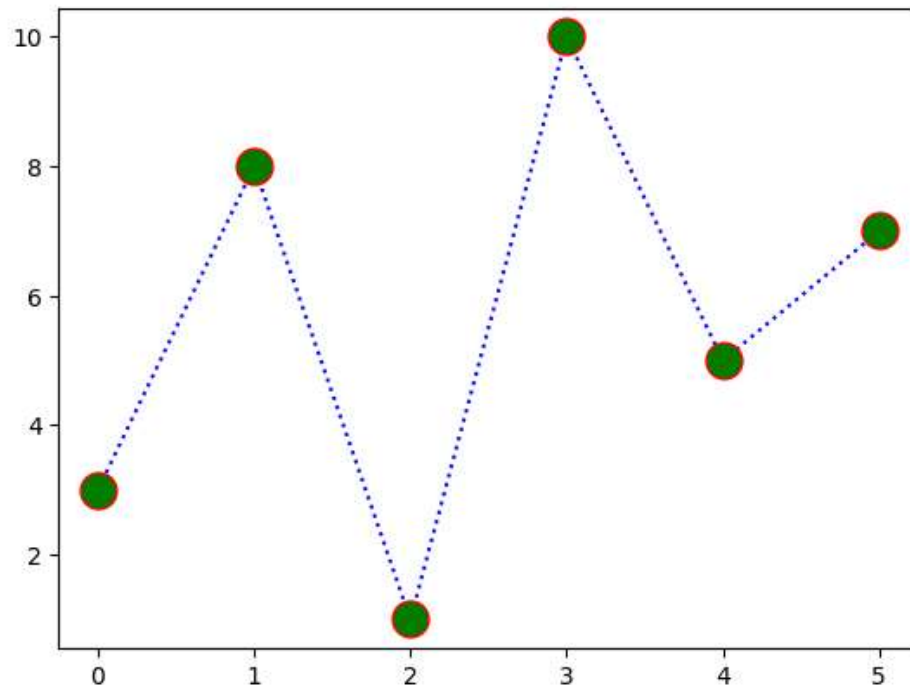


✓ Set Marker Size

#set marker size(ms), marker edge color(mec), marker color(mfc)

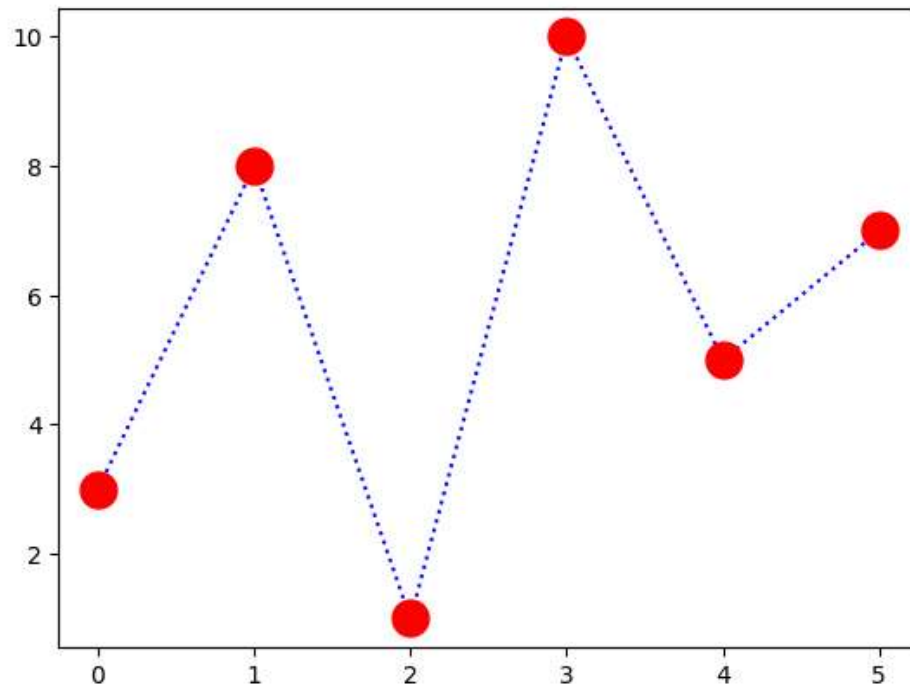
```
ypoints = np.array([3, 8, 1, 10, 5, 7])
```

```
plt.plot(ypoints,'o:b',ms=15, mec='r',mfc='g')  
plt.show()
```



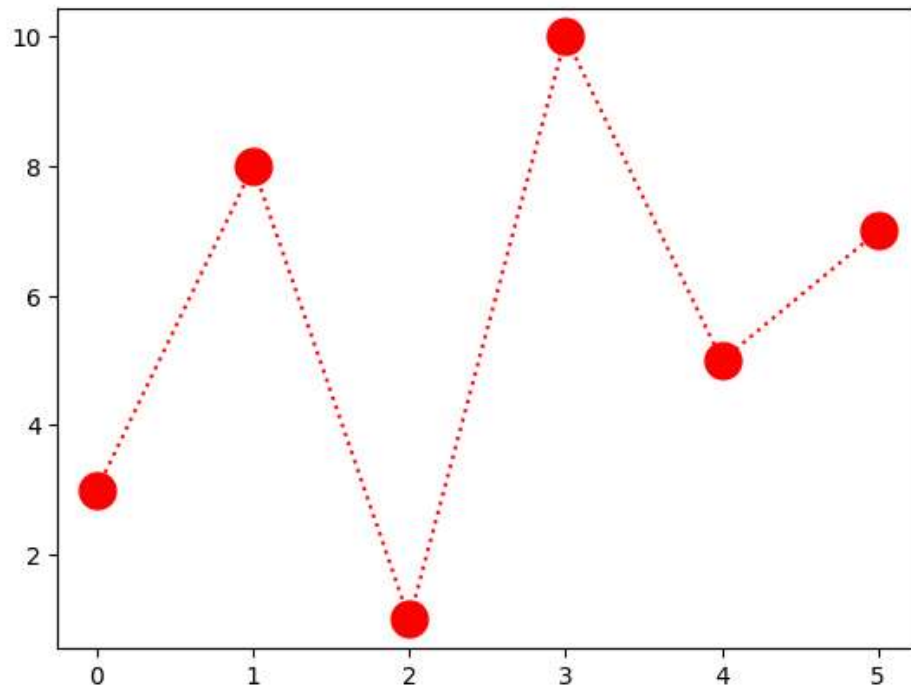
```
#set marker size(ms), marker edge color(mec), marker color(mfc)
ypoints = np.array([3, 8, 1, 10, 5, 7])

plt.plot(ypoints,'o:b',ms=15, mec='r',mfc='r')
plt.show()
```



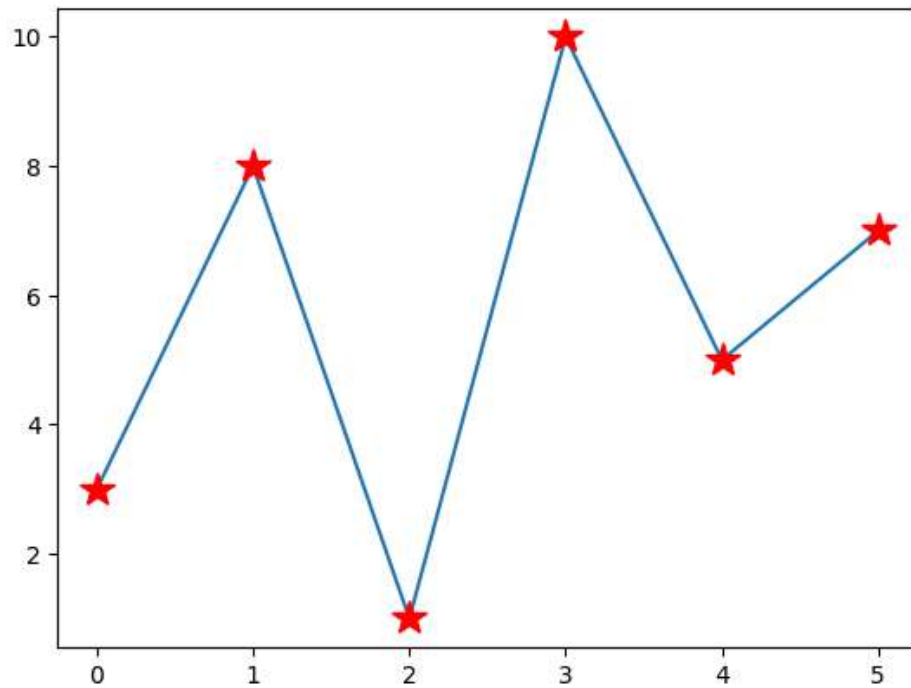
```
#set marker size(ms), marker edge color(mec), marker color(mfc)
ypoints = np.array([3, 8, 1, 10, 5, 7])

plt.plot(ypoints,'o:r',ms=15, mec='r',mfc='r')
plt.show()
```

```
#set marker size(ms), marker edge color(mec), marker color(mfc)
ypoints = np.array([3, 8, 1, 10, 5, 7])

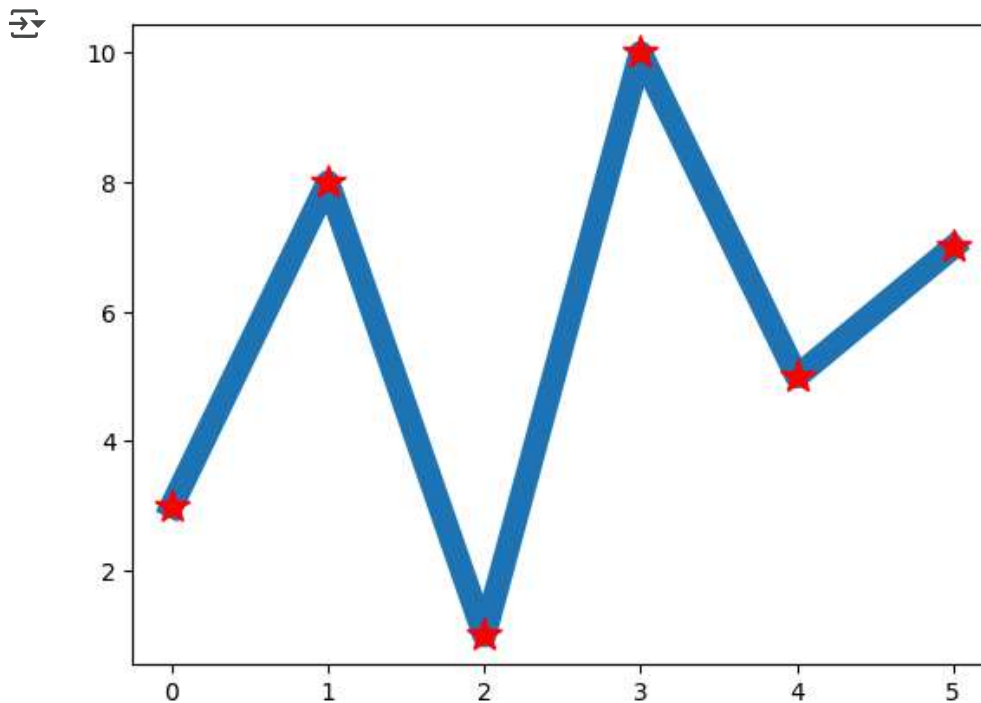
plt.plot(ypoints, marker='*', ms=15, mec='r', mfc='r')
plt.show()
```



✓ set line width

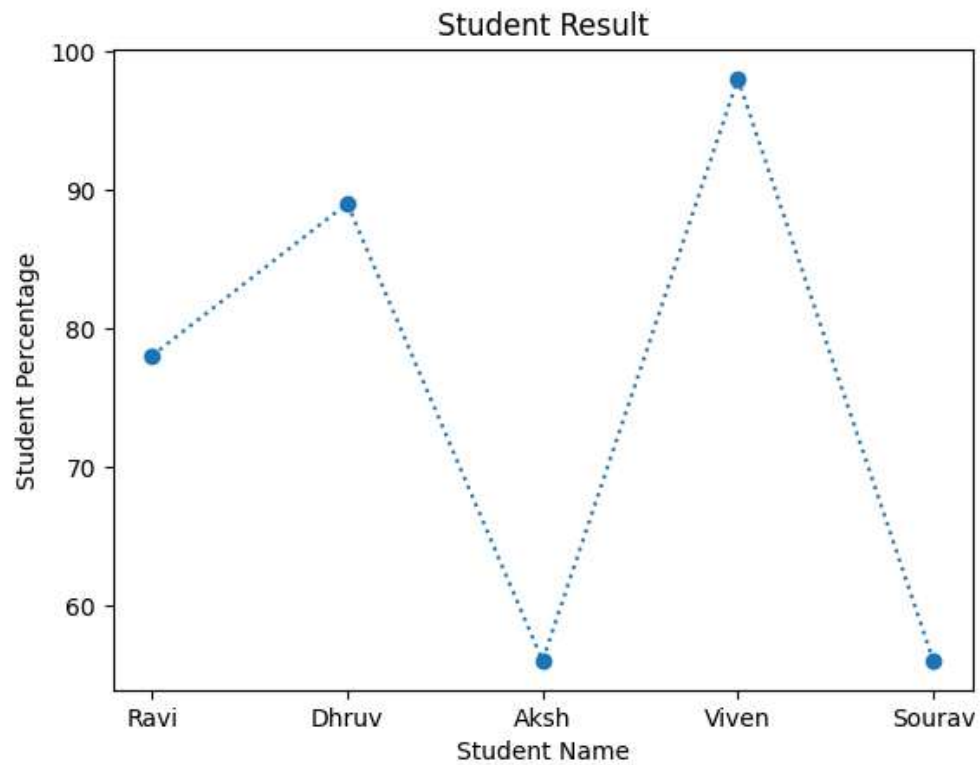
```
ypoints = np.array([3, 8, 1, 10, 5, 7])
```

```
plt.plot(ypoints,marker='*',ms=15, mec='r',mfc='r',lw='10')
plt.show()
```



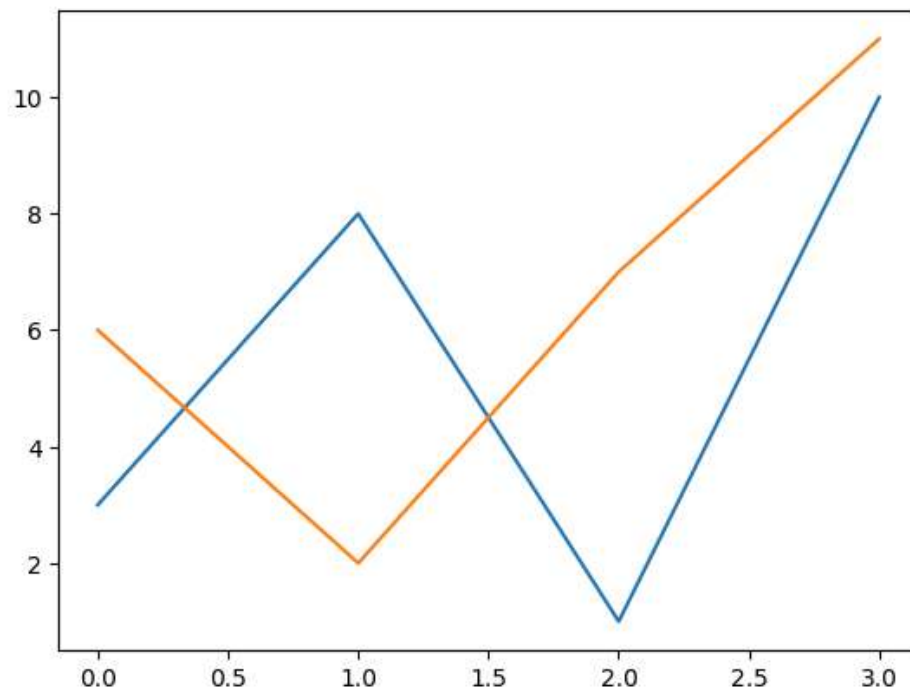
✓ Give title, X label and Y label to plot

```
#Here o for marker and : for line style
name = ['Ravi','Dhruv','Aksh','Viven','Sourav']
percentage = [78,89,56,98,56]
plt.title('Student Result')
plt.xlabel('Student Name')
plt.ylabel('Student Percentage')
plt.plot(name,percentage,'o:')
plt.show()
```



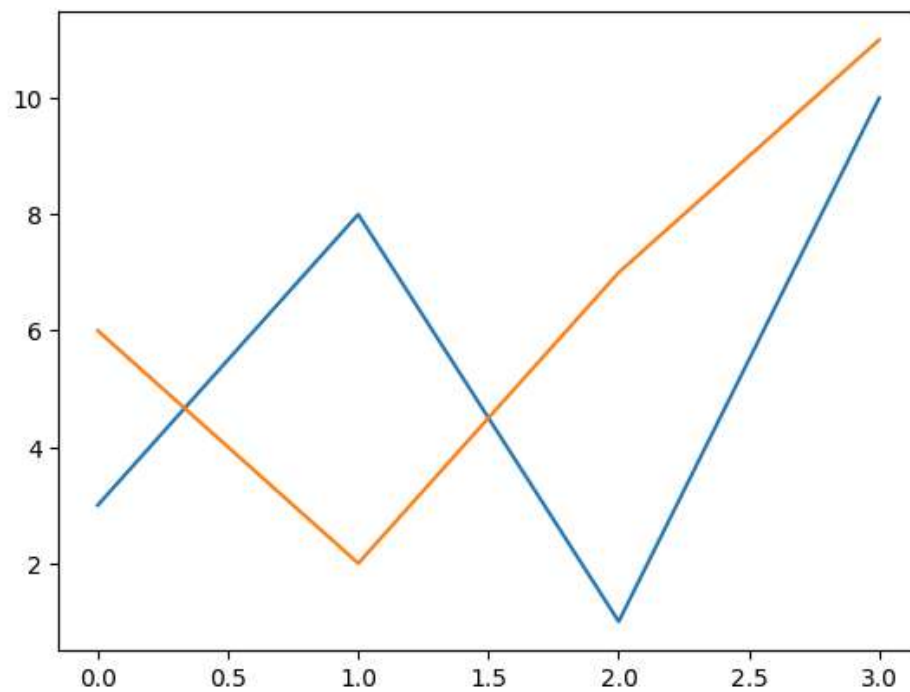
✓ Draw Multiple Plots

```
x1 = np.array([0, 1, 2, 3])
y1 = np.array([3, 8, 1, 10])
x2 = np.array([0, 1, 2, 3])
y2 = np.array([6, 2, 7, 11])
plt.plot(x1,y1)
plt.plot(x2, y2)
plt.show()
```



```
x1 = np.array([0, 1, 2, 3])  
y1 = np.array([3, 8, 1, 10])  
x2 = np.array([0, 1, 2, 3])  
y2 = np.array([6, 2, 7, 11])
```

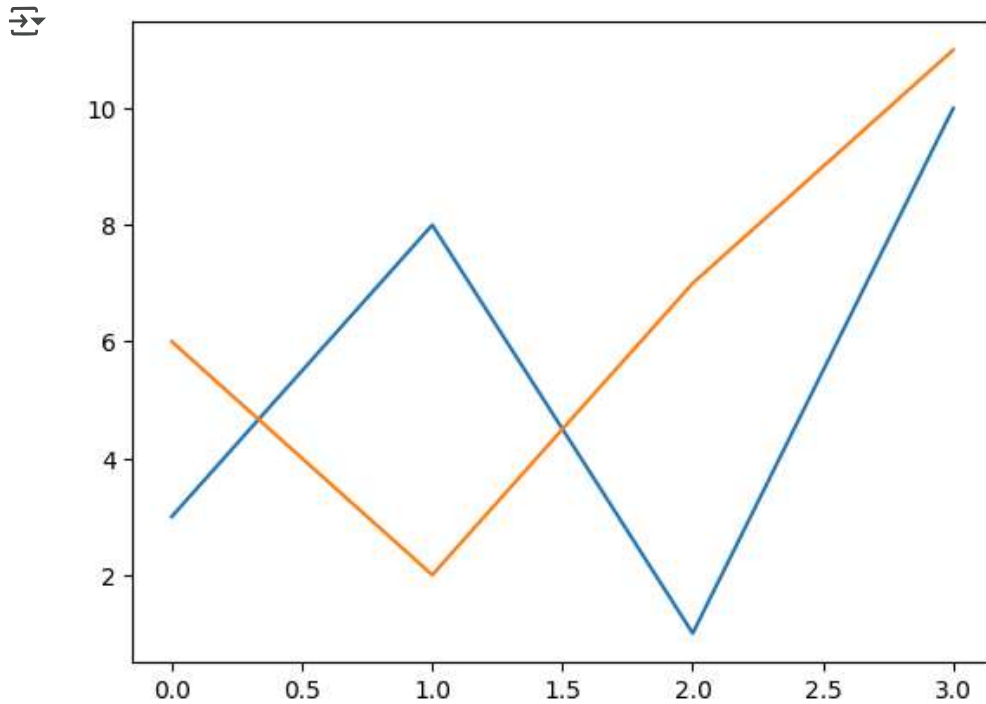
```
plt.plot(x1, y1, x2, y2)  
plt.show()
```



✓ Multiple Plots with default value of x

```
y1 = np.array([3, 8, 1, 10])
y2 = np.array([6, 2, 7, 11])
```

```
plt.plot(y1)
plt.plot(y2)
plt.show()
```



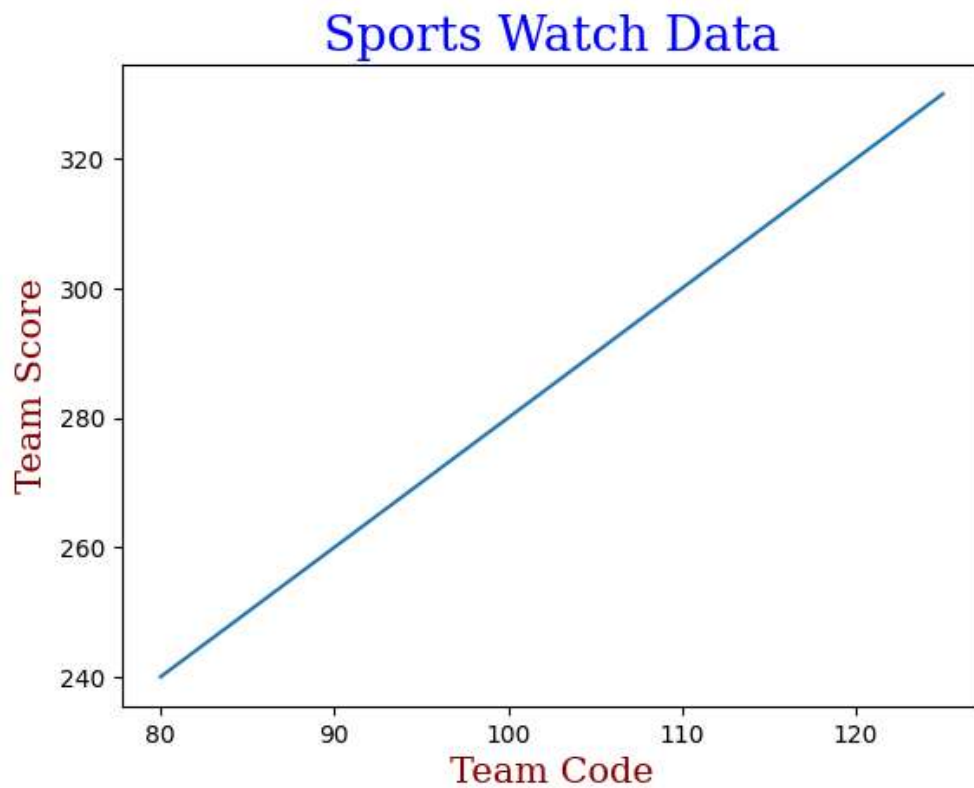
✓ Give styling to label,title

```
x = np.array([80, 85, 90, 95, 100, 105, 110, 115, 120, 125])
y = np.array([240, 250, 260, 270, 280, 290, 300, 310, 320, 330])
```

```
font1 = {'family':'serif','color':'blue','size':20}
font2 = {'family':'serif','color':'darkred','size':15}
```

```
plt.title("Sports Watch Data", fontdict = font1)
plt.xlabel("Team Code", fontdict = font2)
plt.ylabel("Team Score", fontdict = font2)
```

```
plt.plot(x, y)
plt.show()
```

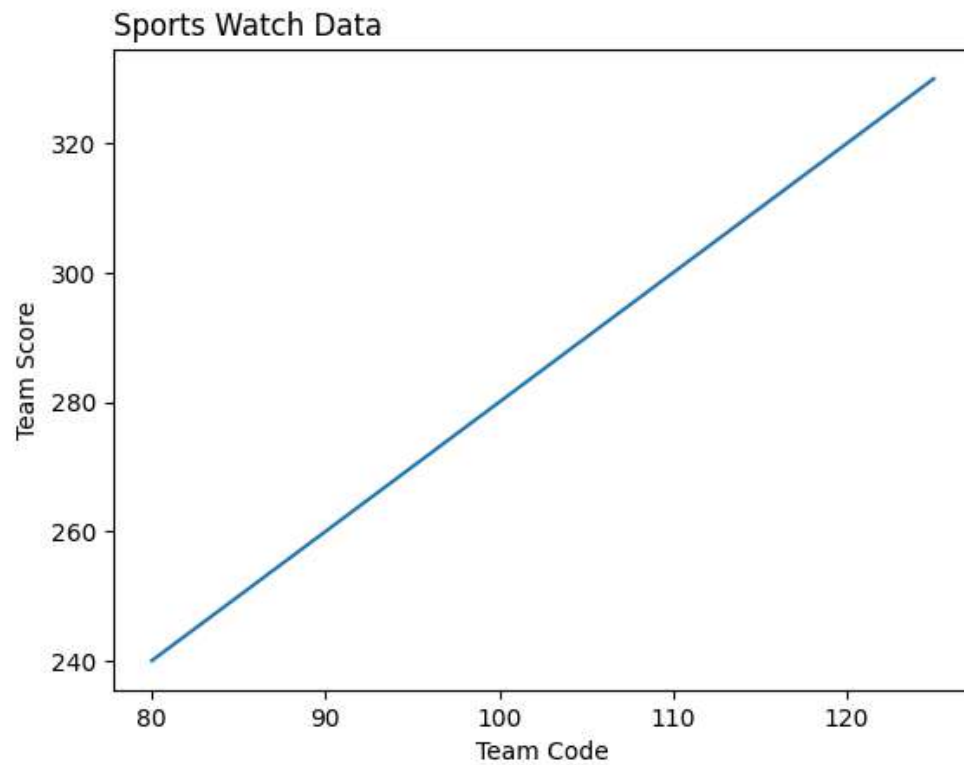


✓ Text alignment

```
x = np.array([80, 85, 90, 95, 100, 105, 110, 115, 120, 125])
y = np.array([240, 250, 260, 270, 280, 290, 300, 310, 320, 330])

plt.title("Sports Watch Data", loc = 'left')
plt.xlabel("Team Code")
plt.ylabel("Team Score")

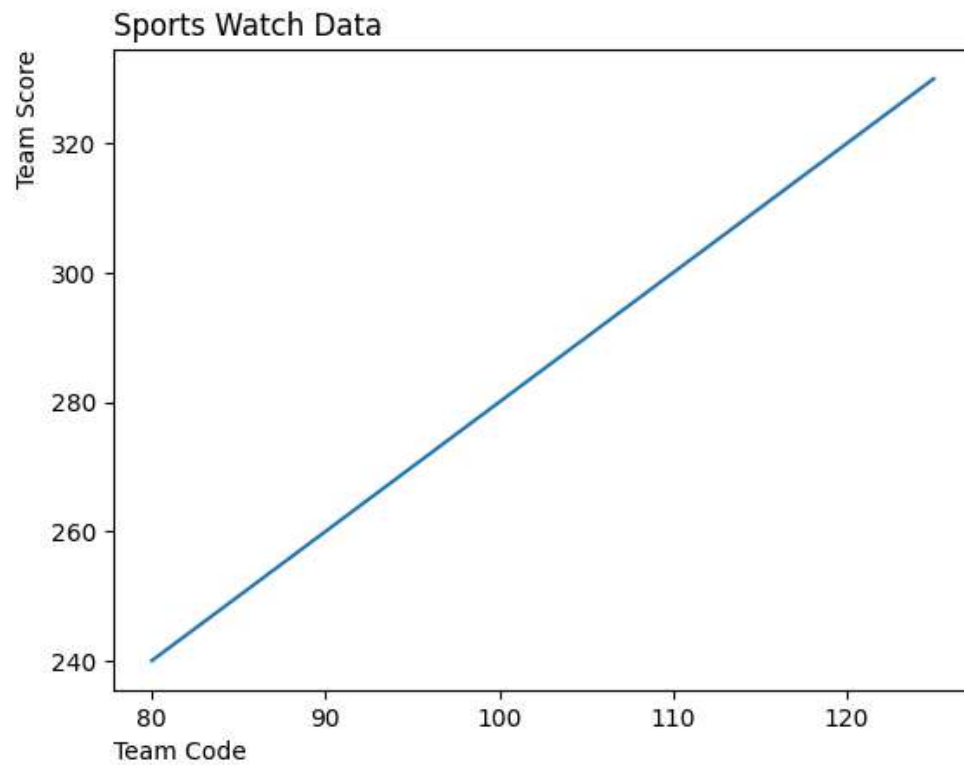
plt.plot(x, y)
plt.show()
```



```
x = np.array([80, 85, 90, 95, 100, 105, 110, 115, 120, 125])
y = np.array([240, 250, 260, 270, 280, 290, 300, 310, 320, 330])
```

```
plt.title("Sports Watch Data", loc = 'left')
plt.xlabel("Team Code", loc = 'left')
plt.ylabel("Team Score",loc = 'top')
```

```
plt.plot(x, y)
plt.show()
```

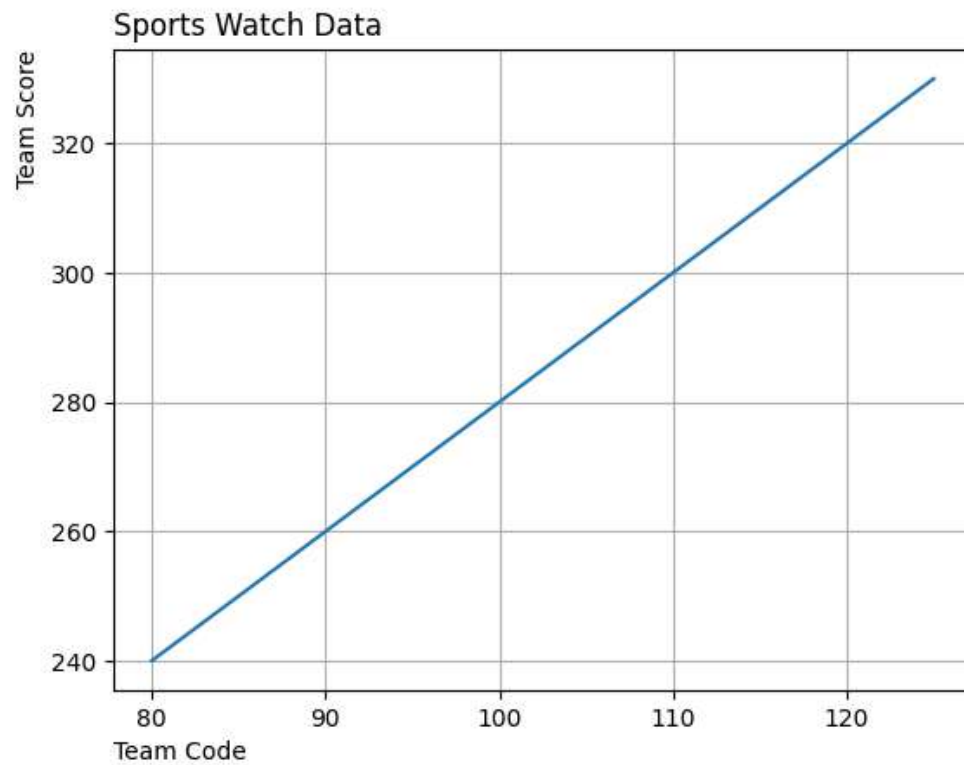


✓ Create Grids

```
x = np.array([80, 85, 90, 95, 100, 105, 110, 115, 120, 125])
y = np.array([240, 250, 260, 270, 280, 290, 300, 310, 320, 330])
```

```
plt.title("Sports Watch Data", loc = 'left')
plt.xlabel("Team Code", loc = 'left')
plt.ylabel("Team Score", loc = 'top')
```

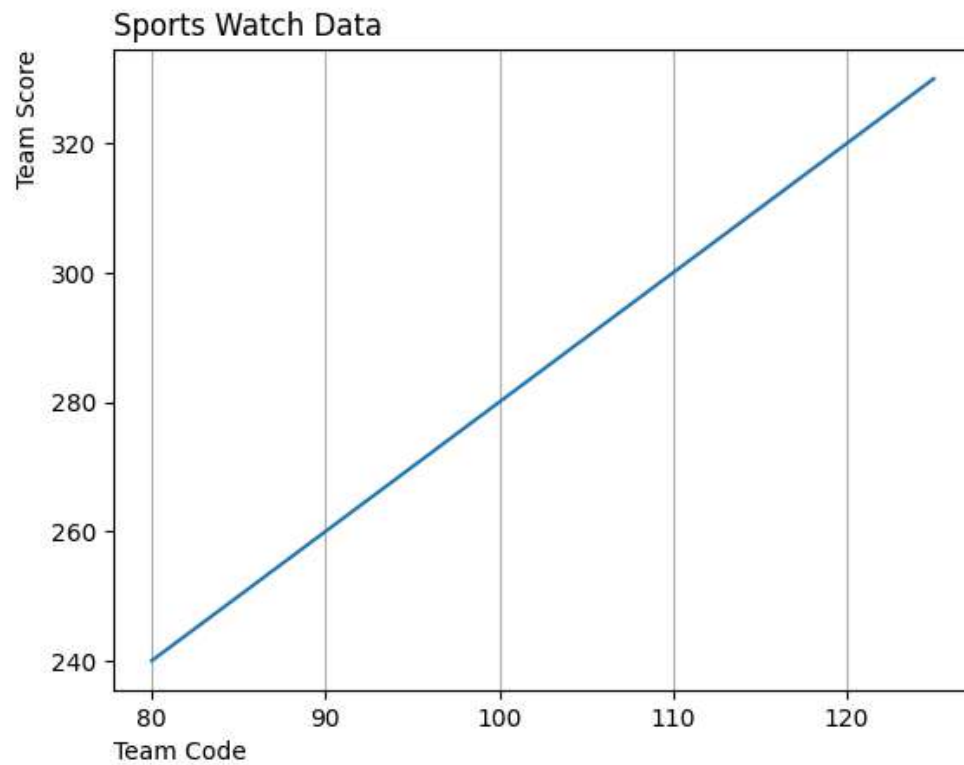
```
plt.plot(x, y)
plt.grid()
plt.show()
```

```
x = np.array([80, 85, 90, 95, 100, 105, 110, 115, 120, 125])
y = np.array([240, 250, 260, 270, 280, 290, 300, 310, 320, 330])
```

```
plt.title("Sports Watch Data", loc = 'left')
plt.xlabel("Team Code", loc = 'left')
plt.ylabel("Team Score",loc = 'top')
```

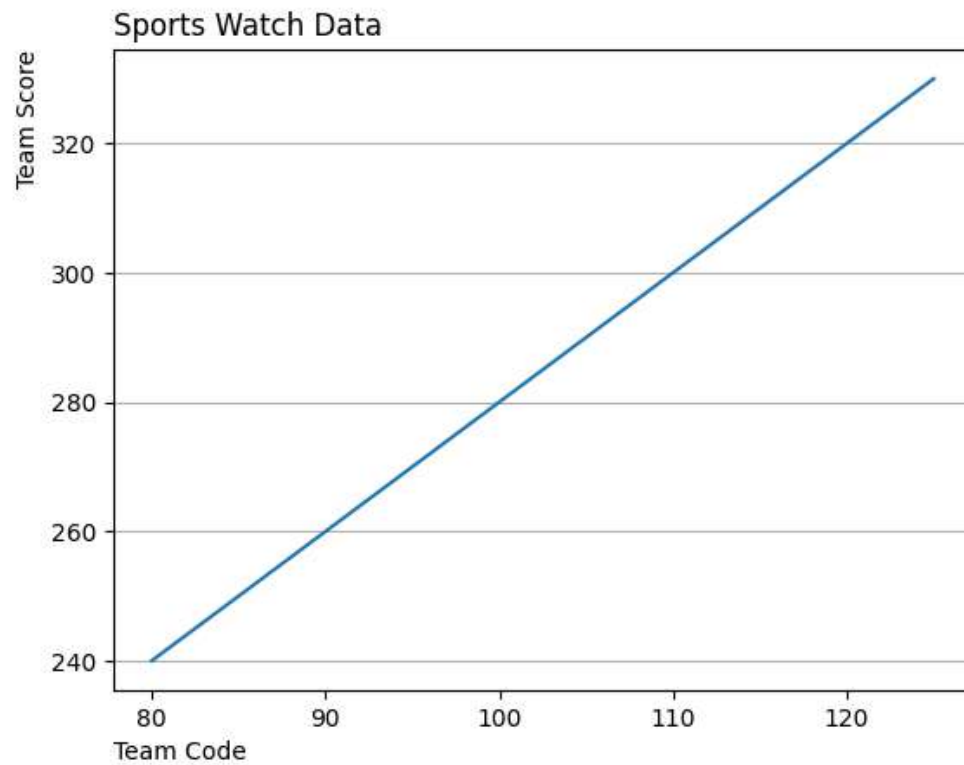
```
plt.plot(x, y)
plt.grid(axis='x')
plt.show()
```



```
x = np.array([80, 85, 90, 95, 100, 105, 110, 115, 120, 125])
y = np.array([240, 250, 260, 270, 280, 290, 300, 310, 320, 330])
```

```
plt.title("Sports Watch Data", loc = 'left')
plt.xlabel("Team Code", loc = 'left')
plt.ylabel("Team Score",loc = 'top')
```

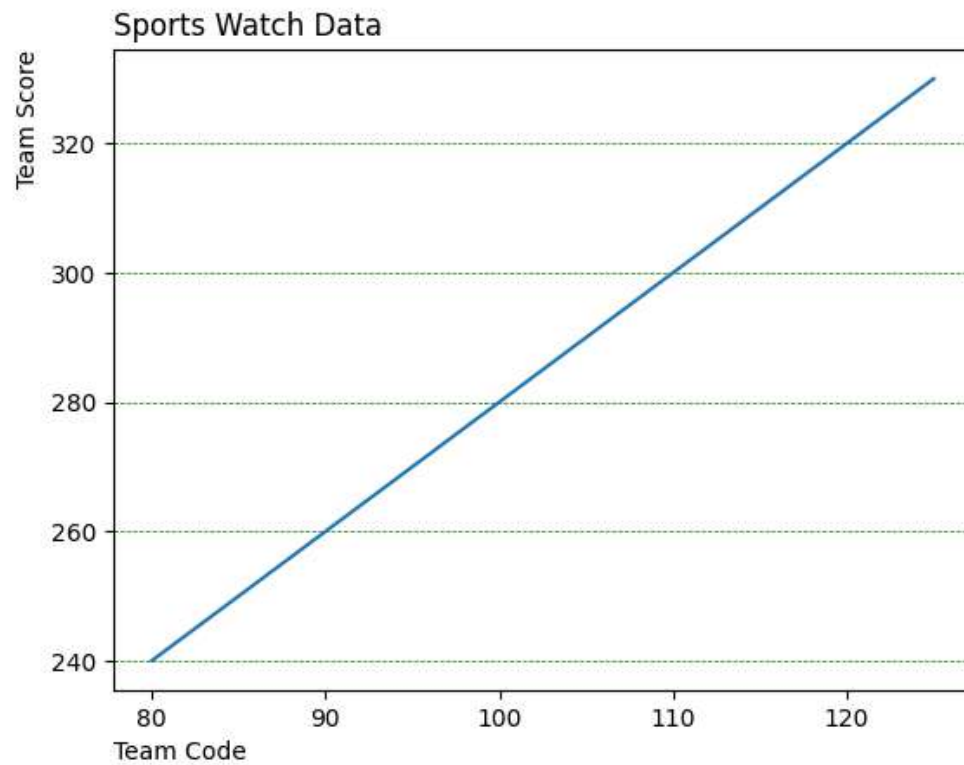
```
plt.plot(x, y)
plt.grid(axis='y')
plt.show()
```



```
x = np.array([80, 85, 90, 95, 100, 105, 110, 115, 120, 125])
y = np.array([240, 250, 260, 270, 280, 290, 300, 310, 320, 330])

plt.title("Sports Watch Data", loc = 'left')
plt.xlabel("Team Code", loc = 'left')
plt.ylabel("Team Score",loc = 'top')

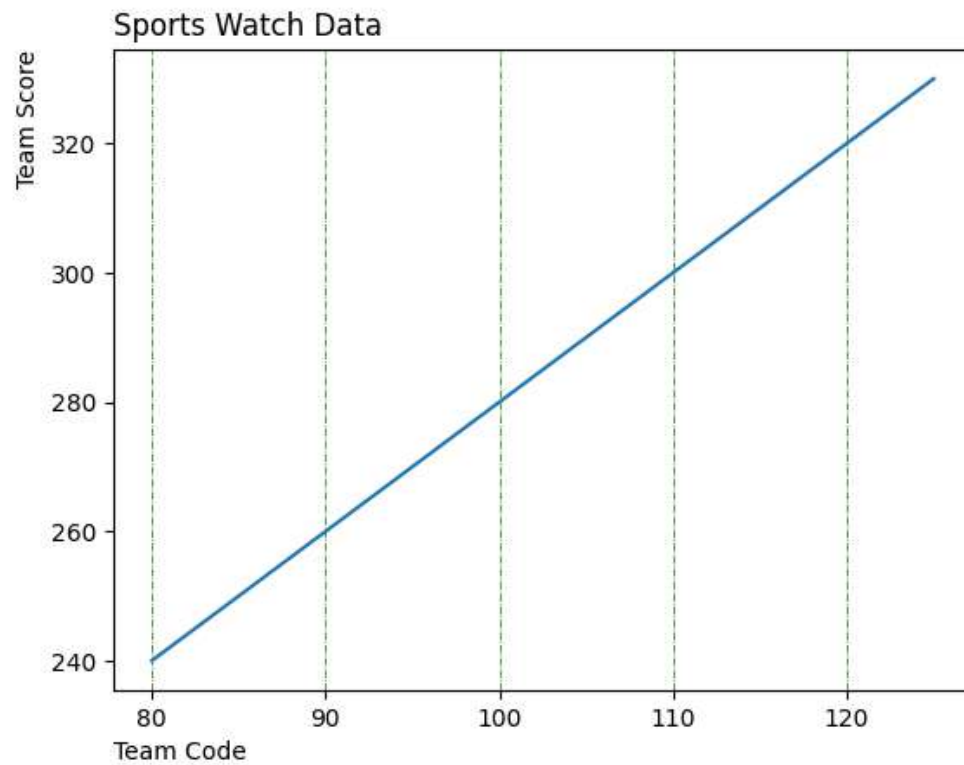
plt.plot(x, y)
plt.grid(axis='y',color = 'green', linestyle = '--', linewidth = 0.5)
plt.show()
```



```
x = np.array([80, 85, 90, 95, 100, 105, 110, 115, 120, 125])
y = np.array([240, 250, 260, 270, 280, 290, 300, 310, 320, 330])

plt.title("Sports Watch Data", loc = 'left')
plt.xlabel("Team Code", loc = 'left')
plt.ylabel("Team Score",loc = 'top')

plt.plot(x, y)
plt.grid(axis='x',color = 'green', linestyle = '-.', linewidth = 0.5)
plt.show()
```



✓ Generate Subplots

`plt.subplot(2, 1, 1)` : first: row, second: column, third: preference

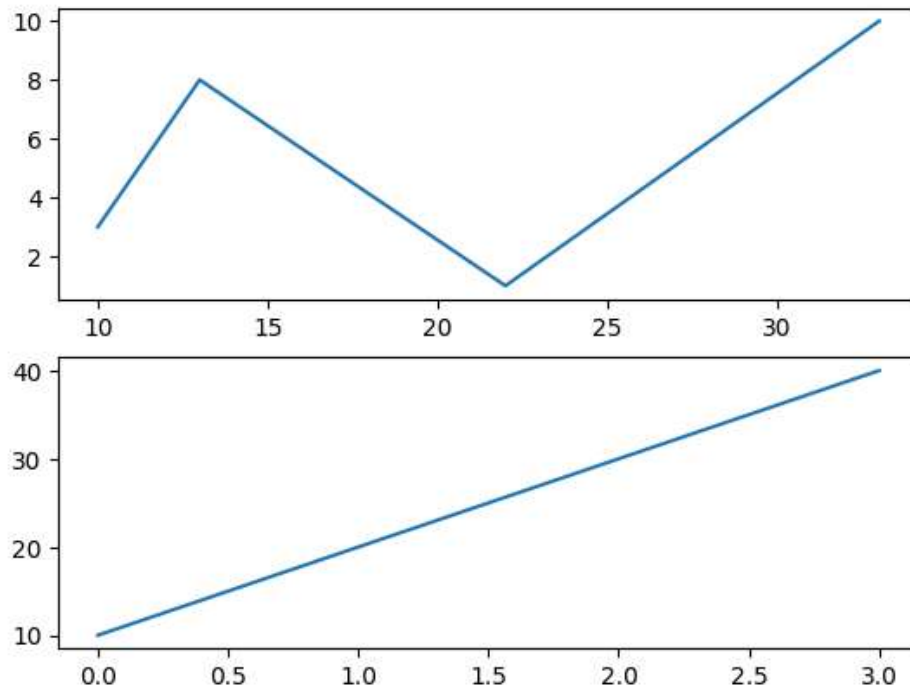
```
#plot 1:  
x = np.array([10, 13, 22, 33])  
y = np.array([3, 8, 1, 10])
```

```
plt.subplot(2, 1, 1)  
plt.plot(x,y)
```

```
#plot 2:  
x = np.array([0, 1, 2, 3])  
y = np.array([10, 20, 30, 40])
```

```
plt.subplot(2, 1, 2)  
plt.plot(x,y)
```

```
plt.show()
```



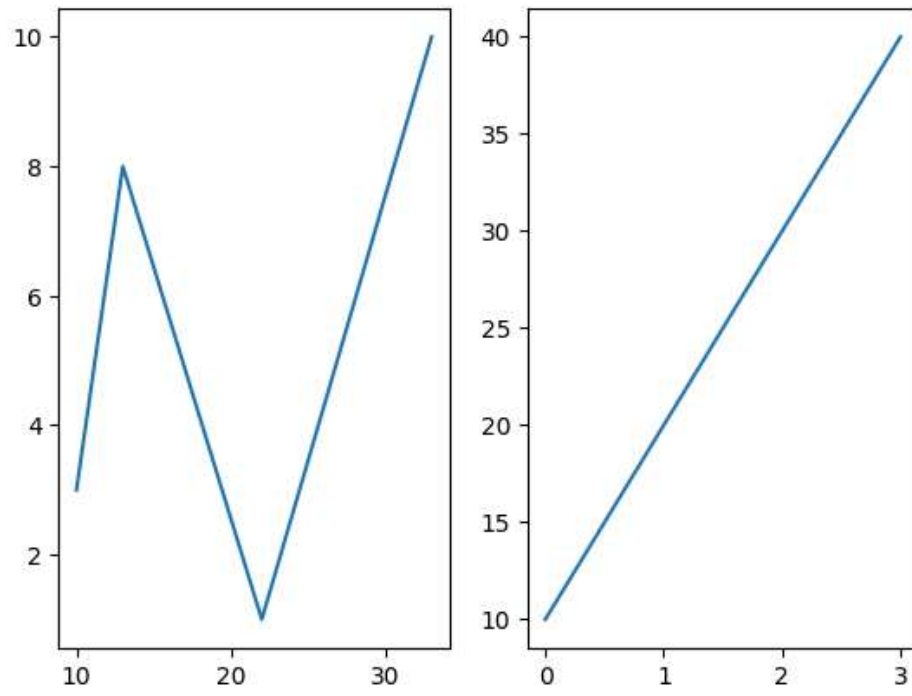
```
#plot 1:
x = np.array([10, 13, 22, 33])
y = np.array([3, 8, 1, 10])

plt.subplot(1, 2, 1)
plt.plot(x,y)

#plot 2:
x = np.array([0, 1, 2, 3])
y = np.array([10, 20, 30, 40])

plt.subplot(1, 2, 2)
plt.plot(x,y)

plt.show()
```



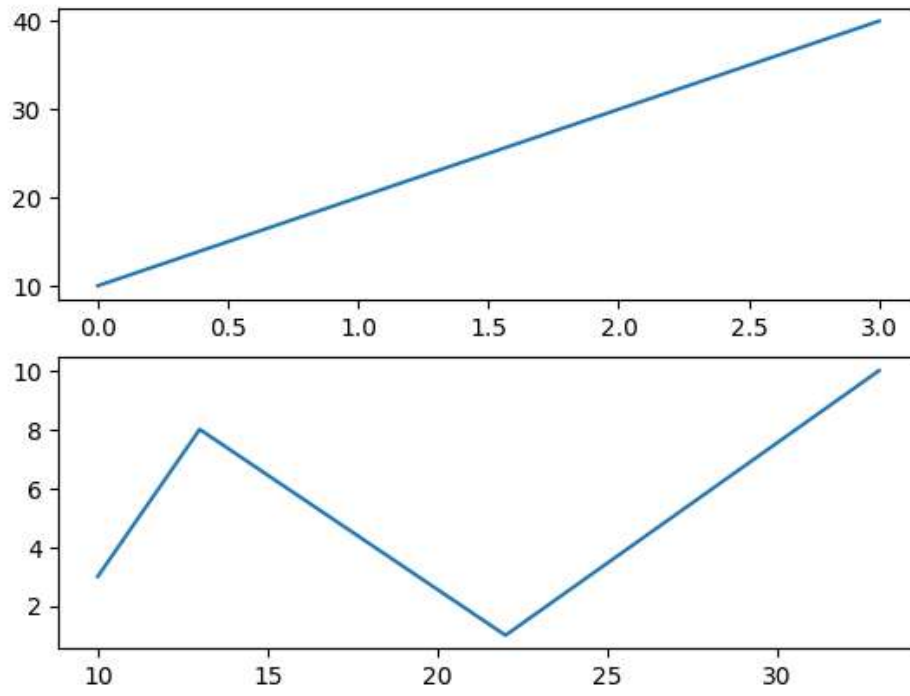
```
#plot 1:  
x = np.array([10, 13, 22, 33])  
y = np.array([3, 8, 1, 10])
```

```
plt.subplot(2, 1, 2)  
plt.plot(x,y)
```

```
#plot 2:  
x = np.array([0, 1, 2, 3])  
y = np.array([10, 20, 30, 40])
```

```
plt.subplot(2, 1, 1)  
plt.plot(x,y)
```

```
plt.show()
```



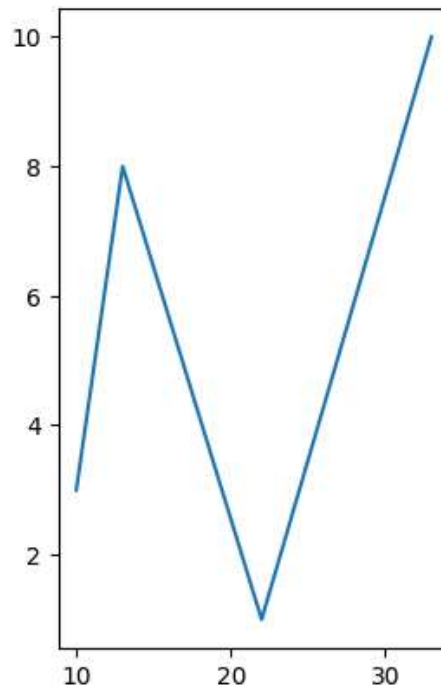
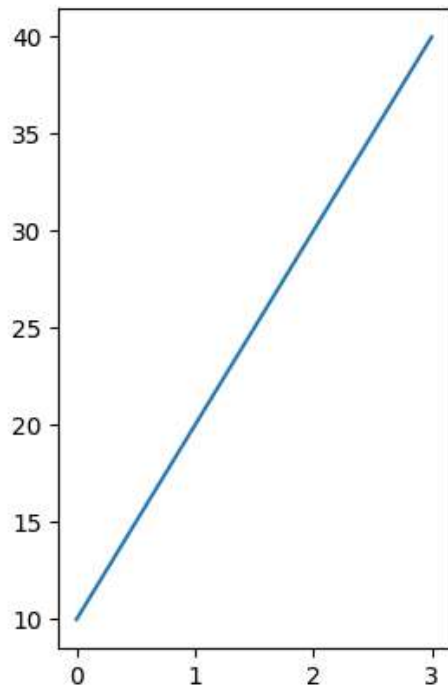
```
#plot 1:  
x = np.array([10, 13, 22, 33])  
y = np.array([3, 8, 1, 10])
```

```
plt.subplot(1, 2, 2)  
plt.plot(x,y)
```

```
#plot 2:  
x = np.array([0, 1, 2, 3])  
y = np.array([10, 20, 30, 40])
```

```
plt.subplot(1, 2, 1)  
plt.plot(x,y)
```

```
plt.show()
```

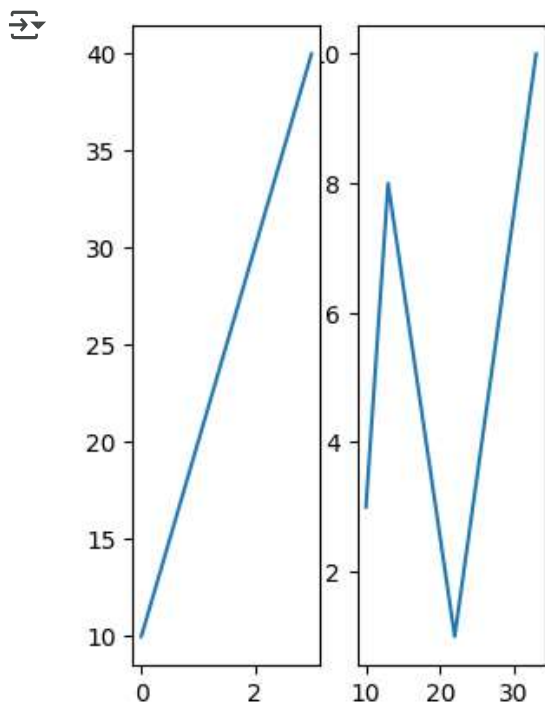
```
#plot 1:  
x = np.array([10, 13, 22, 33])  
y = np.array([3, 8, 1, 10])
```

```
plt.subplot(1, 4, 2)  
plt.plot(x,y)
```

```
#plot 2:  
x = np.array([0, 1, 2, 3])  
y = np.array([10, 20, 30, 40])
```

```
plt.subplot(1, 4, 1)  
plt.plot(x,y)
```

```
plt.show()
```

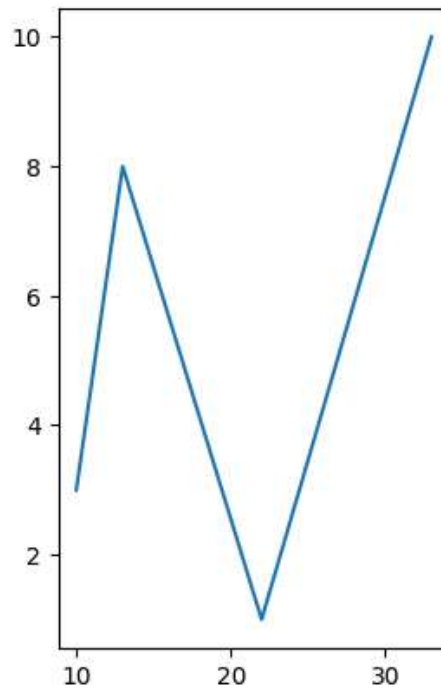
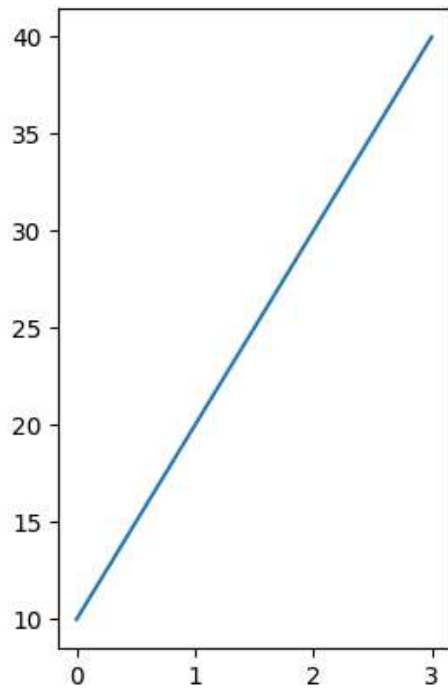


```
#plot 1:  
x = np.array([10, 13, 22, 33])  
y = np.array([3, 8, 1, 10])
```

```
plt.subplot(1, 2, 2)  
plt.plot(x,y)
```

```
#plot 2:  
x = np.array([0, 1, 2, 3])  
y = np.array([10, 20, 30, 40])
```

```
plt.subplot(1, 2, 1)  
plt.plot(x,y)  
plt.show()
```



```
x = np.array([0, 1, 2, 3])
y = np.array([3, 8, 1, 10])
```

```
plt.subplot(2, 3, 1)
plt.plot(x,y)
```

```
x = np.array([0, 1, 2, 3])
y = np.array([10, 20, 30, 40])
```

```
plt.subplot(2, 3, 2)
plt.plot(x,y)
```

```
x = np.array([0, 1, 2, 3])
y = np.array([3, 8, 1, 10])
```

```
plt.subplot(2, 3, 3)
plt.plot(x,y)
```

```
x = np.array([0, 1, 2, 3])
y = np.array([10, 20, 30, 40])
```

```
plt.subplot(2, 3, 4)
plt.plot(x,y)
```

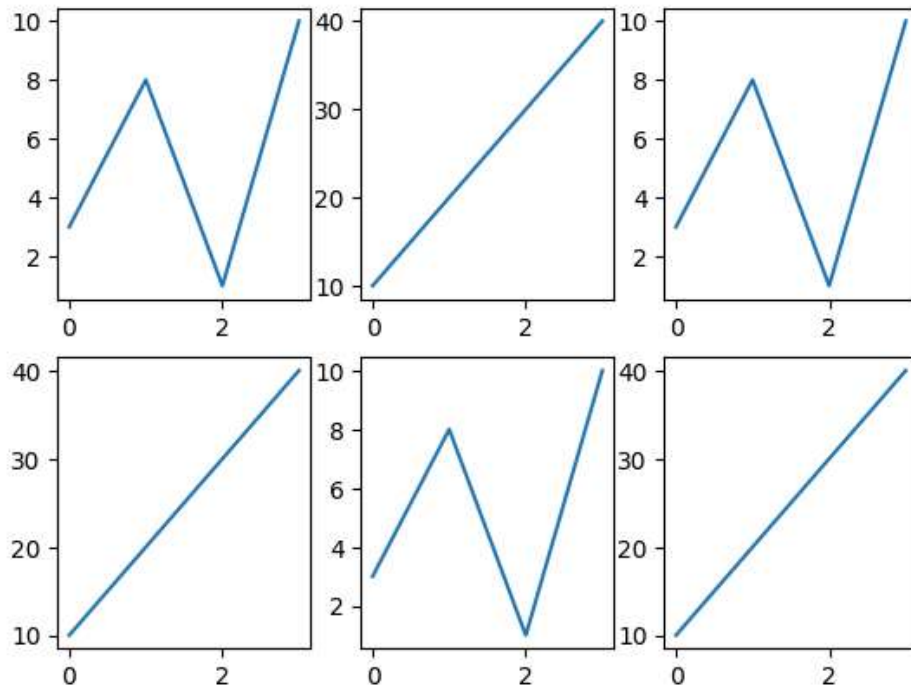
```
x = np.array([0, 1, 2, 3])
y = np.array([3, 8, 1, 10])
```

```
plt.subplot(2, 3, 5)
plt.plot(x,y)
```

```
x = np.array([0, 1, 2, 3])
y = np.array([10, 20, 30, 40])
```

```
plt.subplot(2, 3, 6)
plt.plot(x,y)
```

```
plt.show()
```



✓ Give title to subplot

```
x = np.array([0, 1, 2, 3])
y = np.array([3, 8, 1, 10])
```

```
plt.subplot(2, 3, 1)
plt.title('First Plot')
plt.plot(x,y)
```

```
x = np.array([0, 1, 2, 3])
y = np.array([10, 20, 30, 40])
```

```
plt.subplot(2, 3, 2)
plt.title('Second Plot')
plt.plot(x,y)
```

```
x = np.array([0, 1, 2, 3])
y = np.array([3, 8, 1, 10])
```

```
plt.subplot(2, 3, 3)
plt.title('Third Plot')
plt.plot(x,y)
```

```
x = np.array([0, 1, 2, 3])
y = np.array([10, 20, 30, 40])
```

```
plt.subplot(2, 3, 4)
plt.title('Fourth Plot')
plt.plot(x,y)
```

```
x = np.array([0, 1, 2, 3])
```

```

y = np.array([3, 8, 1, 10])

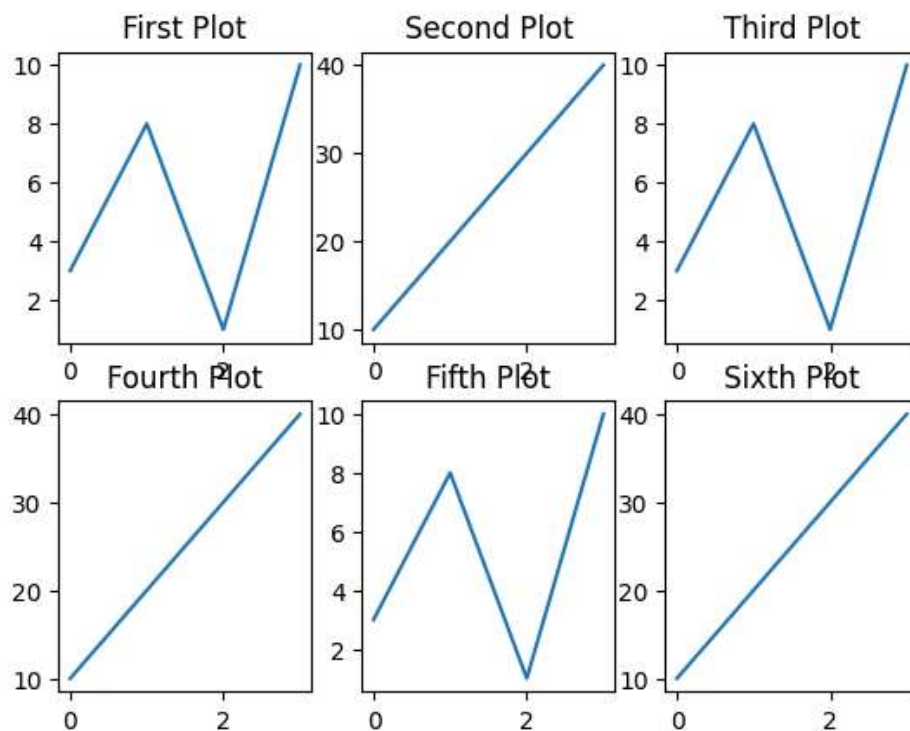
plt.subplot(2, 3, 5)
plt.title('Fifth Plot')
plt.plot(x,y)

x = np.array([0, 1, 2, 3])
y = np.array([10, 20, 30, 40])

plt.subplot(2, 3, 6)
plt.title('Sixth Plot')
plt.plot(x,y)

plt.show()

```



✓ Give super title

```

x = np.array([0, 1, 2, 3])
y = np.array([3, 8, 1, 10])

plt.subplot(2, 3, 1)
plt.title('First Plot')
plt.plot(x,y)

x = np.array([0, 1, 2, 3])
y = np.array([10, 20, 30, 40])

plt.subplot(2, 3, 2)
plt.title('Second Plot')
plt.plot(x,y)

x = np.array([0, 1, 2, 3])
y = np.array([3, 8, 1, 10])

```

```
plt.subplot(2, 3, 3)
plt.title('Third Plot')
plt.plot(x,y)

x = np.array([0, 1, 2, 3])
y = np.array([10, 20, 30, 40])
```

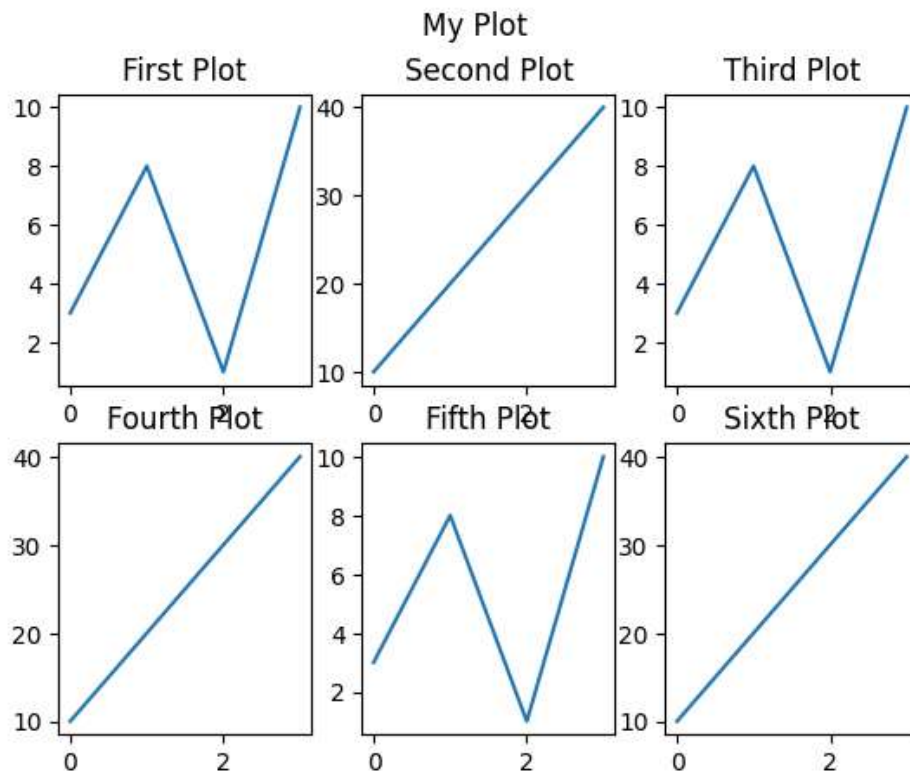
```
plt.subplot(2, 3, 4)
plt.title('Fourth Plot')
plt.plot(x,y)
```

```
x = np.array([0, 1, 2, 3])
y = np.array([3, 8, 1, 10])
```

```
plt.subplot(2, 3, 5)
plt.title('Fifth Plot')
plt.plot(x,y)
```

```
x = np.array([0, 1, 2, 3])
y = np.array([10, 20, 30, 40])
```

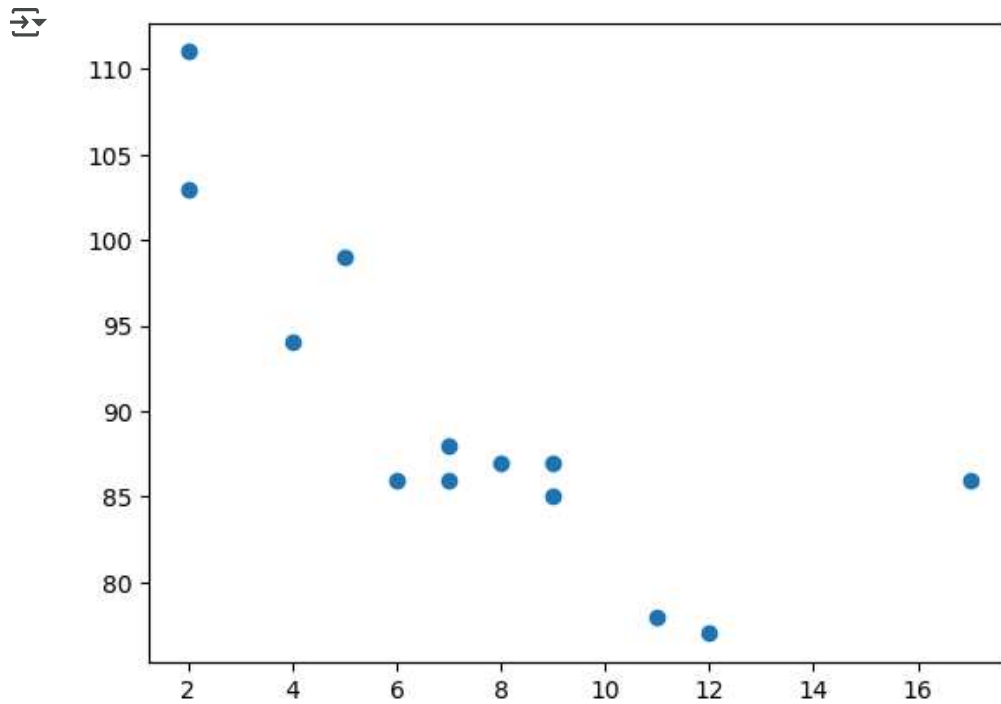
```
plt.subplot(2, 3, 6)
plt.title('Sixth Plot')
plt.plot(x,y)
plt.suptitle('My Plot')
plt.show()
```



✓ Scatter plot

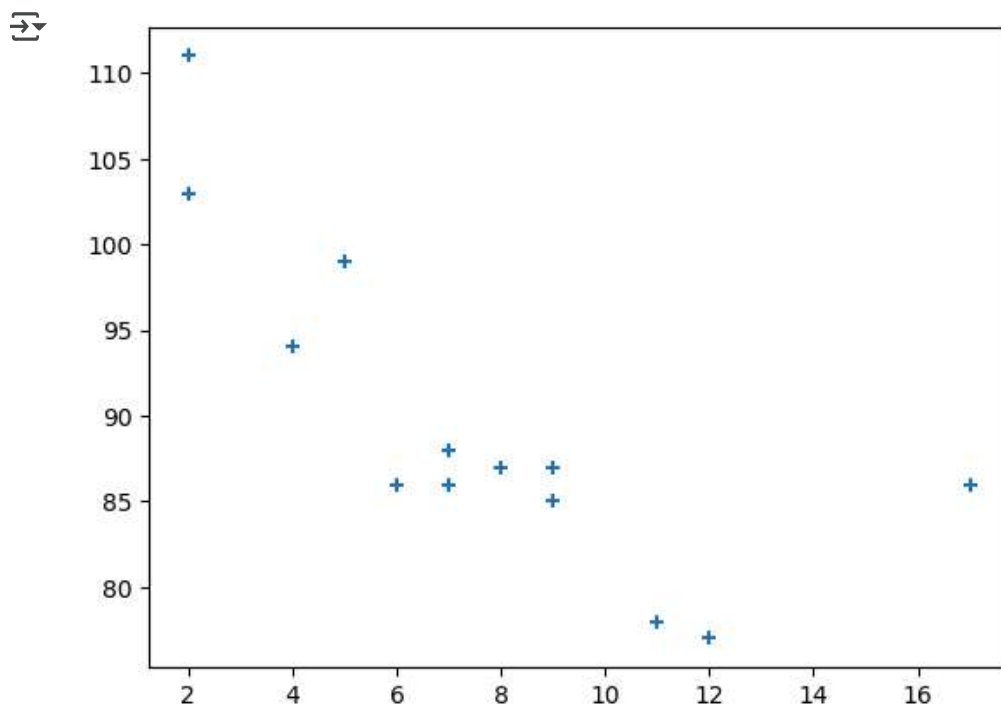
```
x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])
```

```
plt.scatter(x, y)
plt.show()
```



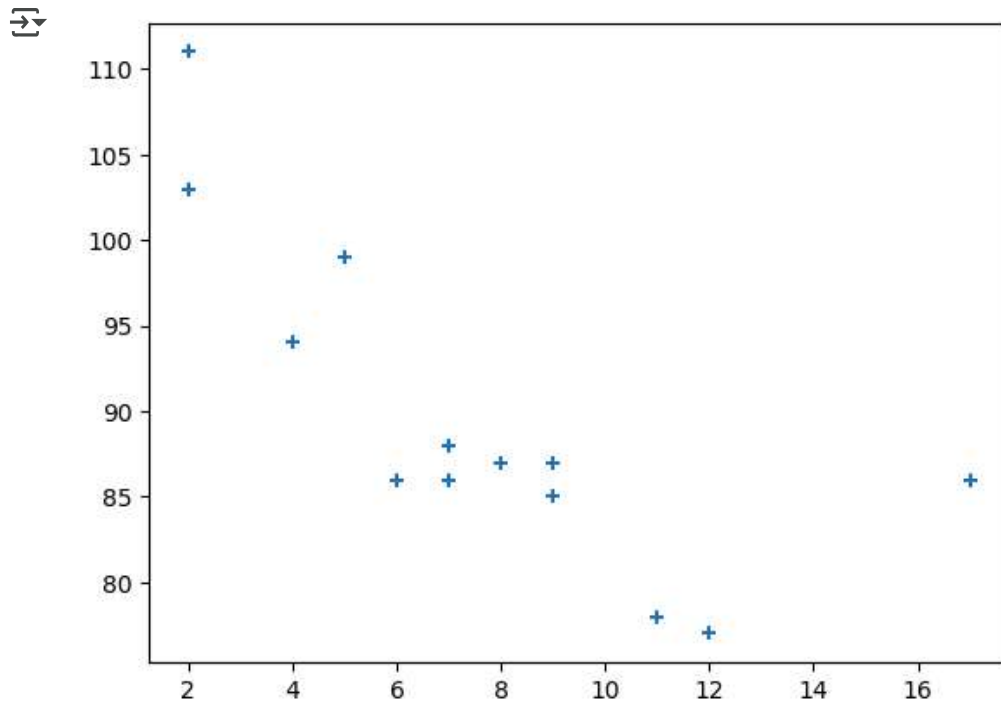
```
x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])
```

```
plt.scatter(x, y, marker='+')
plt.show()
```



```
x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])
```

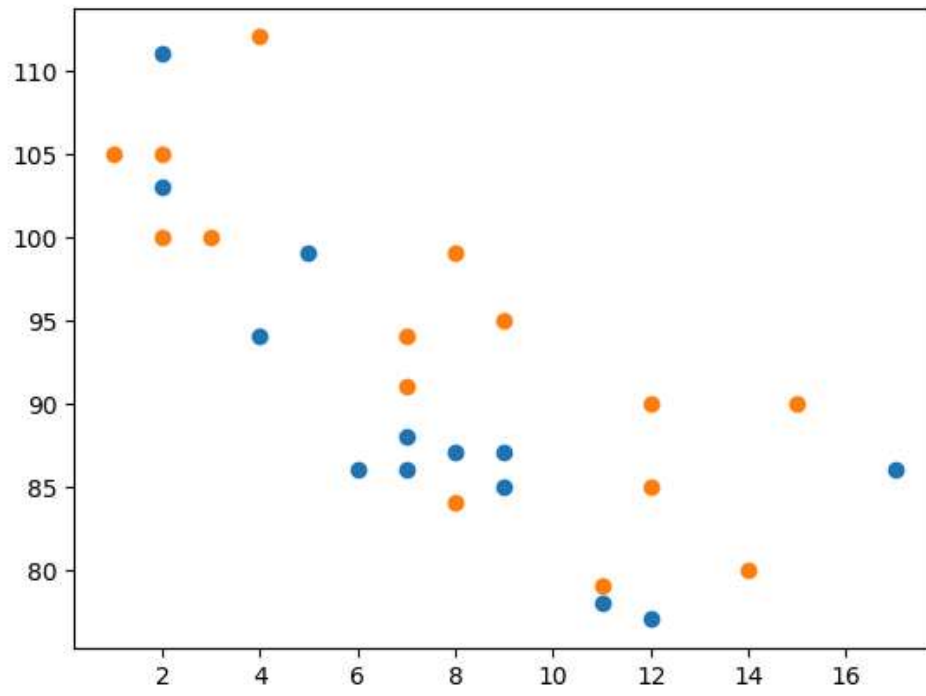
```
plt.scatter(x, y, marker='+')
plt.show()
```



```
#day one, the age and speed of 13 cars:
x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])
plt.scatter(x, y)
```

```
#day two, the age and speed of 15 cars:
x = np.array([2,2,8,1,15,8,12,9,7,3,11,4,7,14,12])
y = np.array([100,105,84,105,90,99,90,95,94,100,79,112,91,80,85])
plt.scatter(x, y)
```

```
plt.show()
```

✓ set color for scatter plot

#day one, the age and speed of 13 cars:

```
x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
```

```
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])
```

```
plt.scatter(x, y,color='r')
```

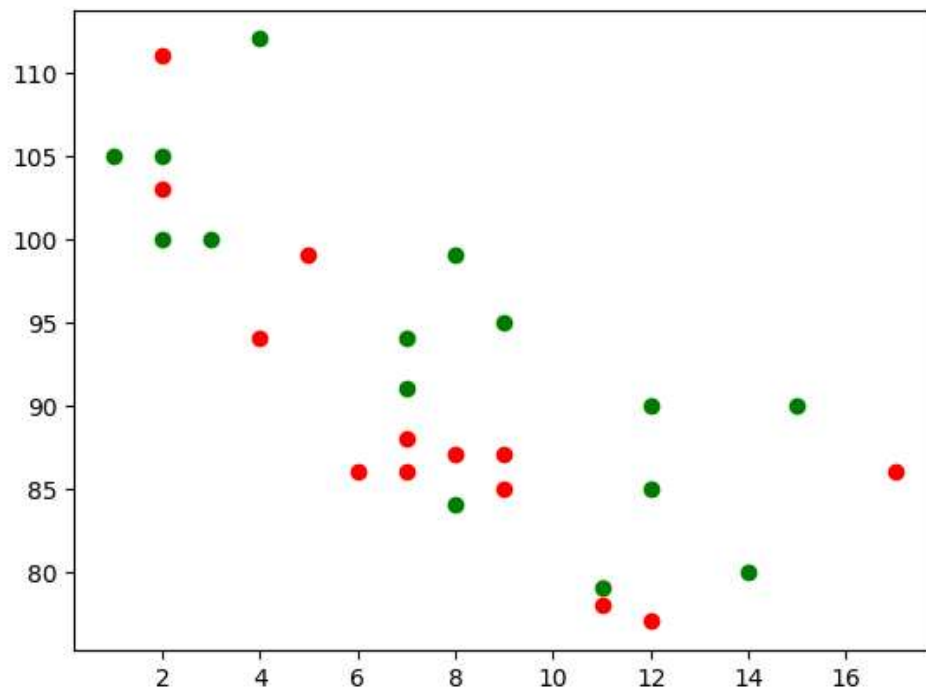
#day two, the age and speed of 15 cars:

```
x = np.array([2,2,8,1,15,8,12,9,7,3,11,4,7,14,12])
```

```
y = np.array([100,105,84,105,90,99,90,95,94,100,79,112,91,80,85])
```

```
plt.scatter(x, y,color='g')
```

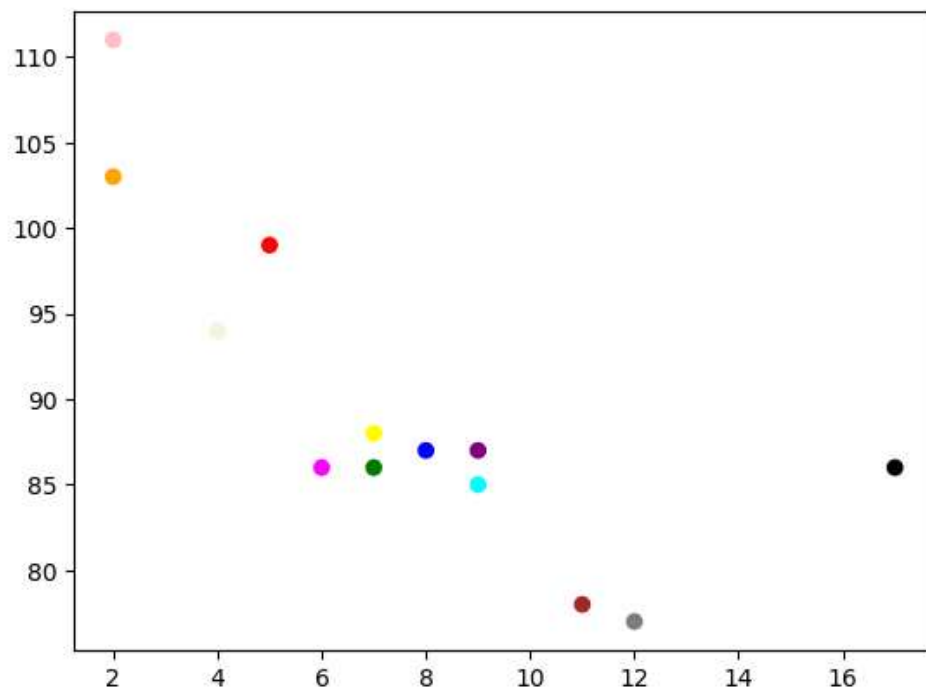
```
plt.show()
```



```
x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])
colors = np.array(["red","green","blue","yellow","pink","black","orange","purple","beige","brown","gray","cyan","magenta"])

plt.scatter(x, y, c=colors)

plt.show()
```



```
#day one, the age and speed of 13 cars:
x = np.array([5,7,8,7])
y = np.array([99,86,87,88])
```

```
plt.scatter(x, y)
```

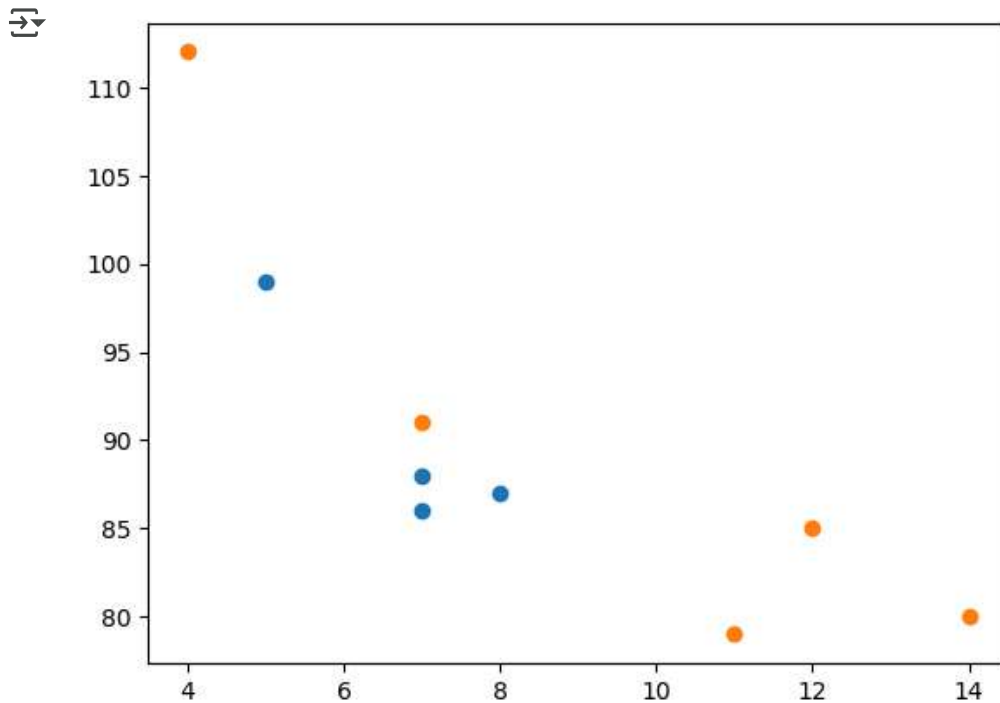
```
#day two, the age and speed of 15 cars:
```

```
x = np.array([11,4,7,14,12])
```

```
y = np.array([79,112,91,80,85])
```

```
plt.scatter(x, y)
```

```
plt.show()
```



```
#day one, the age and speed of 13 cars:
```

```
x = np.array([5,7,8,7])
```

```
y = np.array([99,86,87,88])
```

```
colors = ["red","green","blue","yellow"]
```

```
plt.scatter(x, y, c=colors)
```

```
#day two, the age and speed of 15 cars:
```

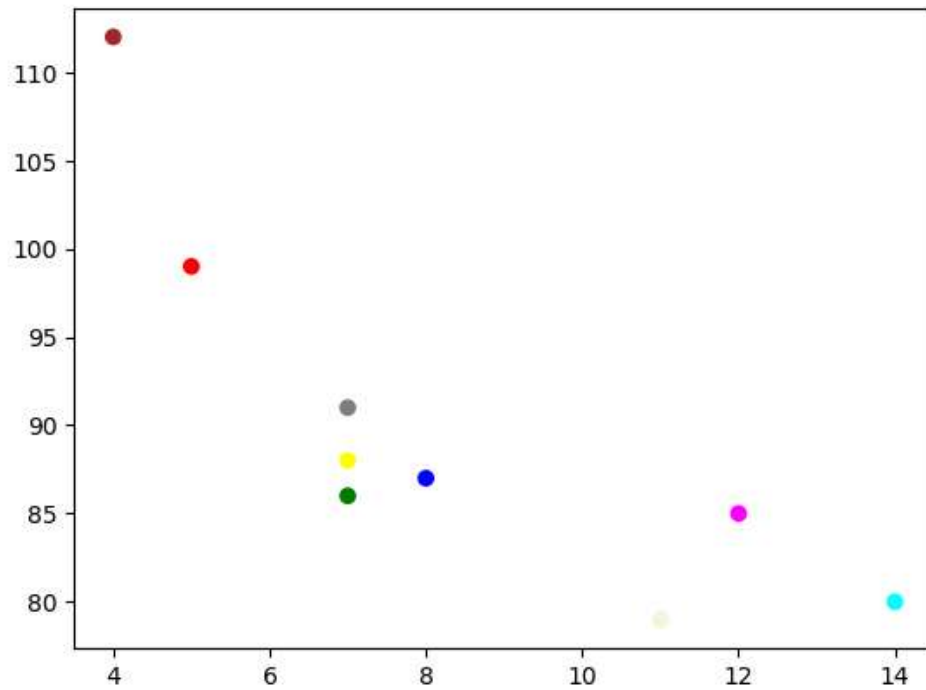
```
x = np.array([11,4,7,14,12])
```

```
y = np.array([79,112,91,80,85])
```

```
colors = ["beige","brown","gray","cyan","magenta"]
```

```
plt.scatter(x, y, c=colors)
```

```
plt.show()
```

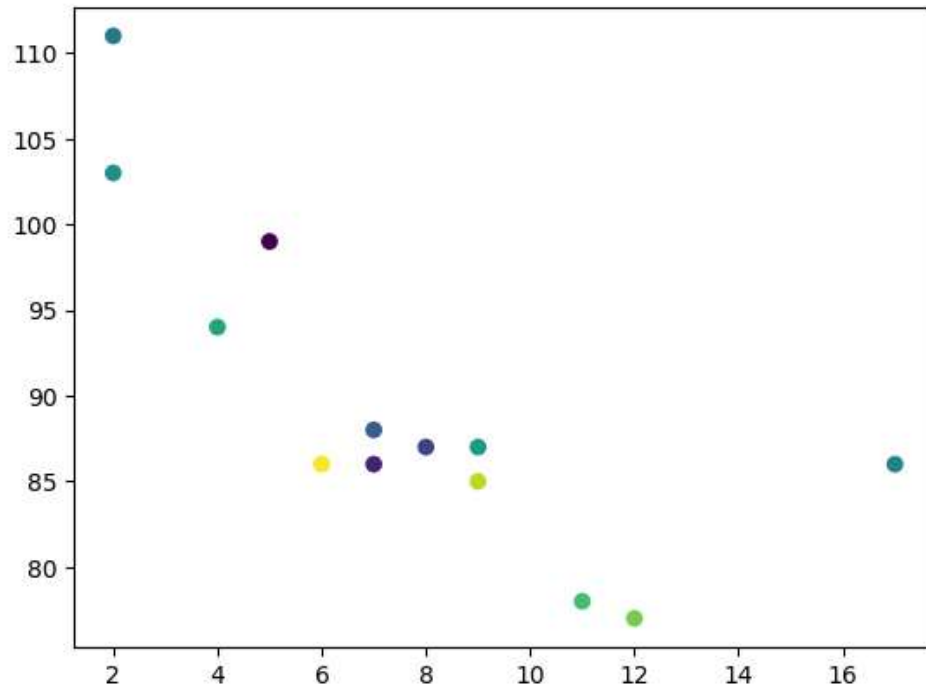


✓ Use color map to give colors to plot

```
x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])
colors = np.array([0, 10, 20, 30, 40, 45, 50, 55, 60, 70, 80, 90, 100])

plt.scatter(x, y, c=colors, cmap='viridis')

plt.show()
```

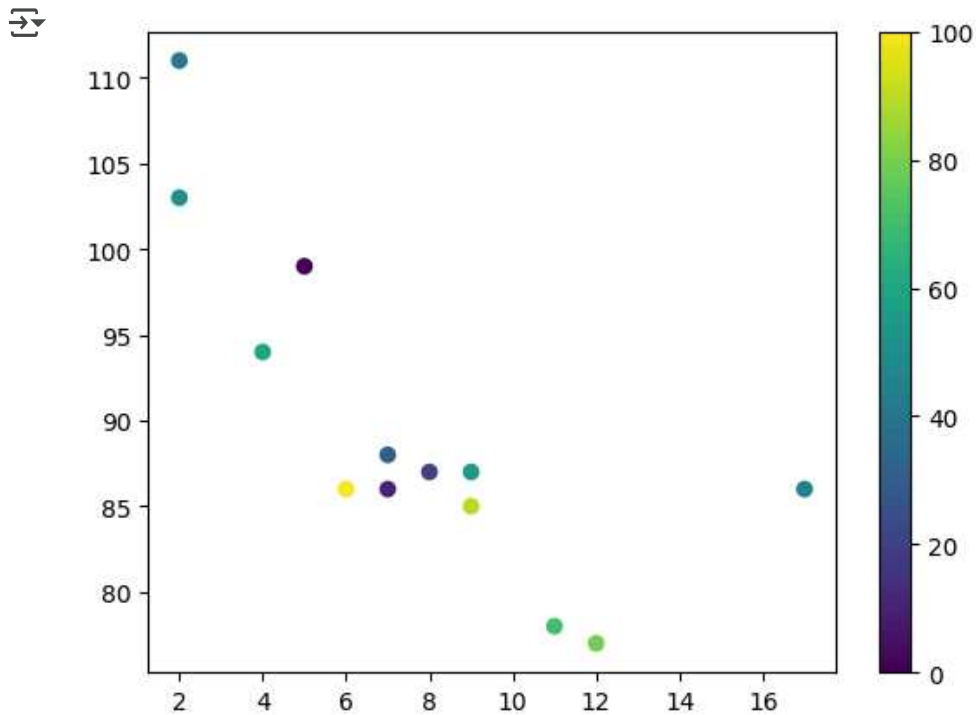


```
x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])
colors = np.array([0, 10, 20, 30, 40, 45, 50, 55, 60, 70, 80, 90, 100])

plt.scatter(x, y, c=colors, cmap='viridis')

plt.colorbar()

plt.show()
```



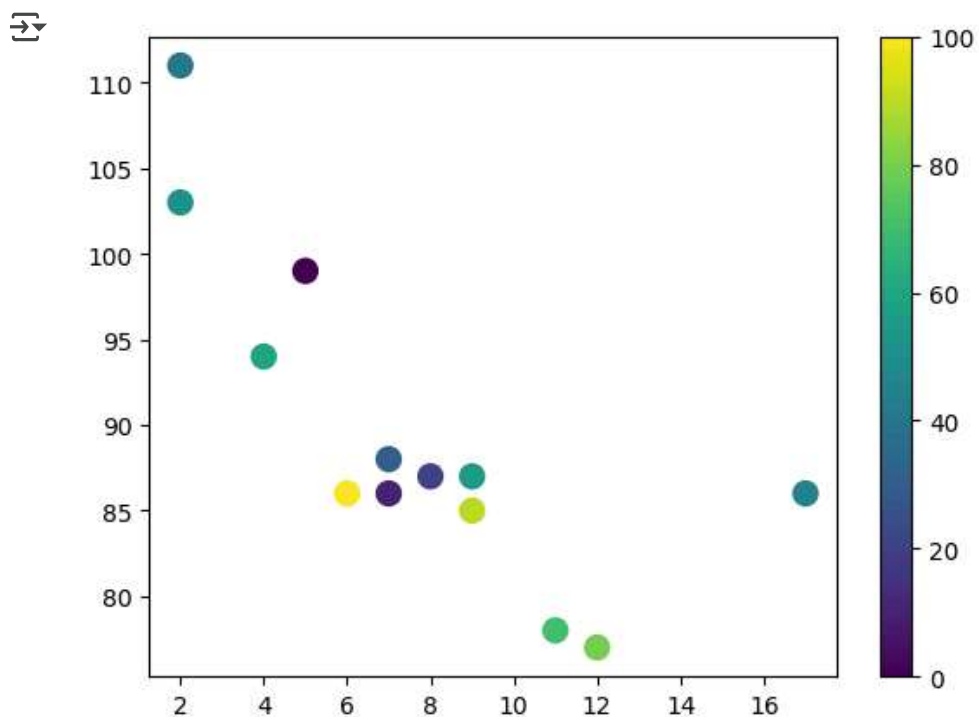
✓ Give size s to dots

```
x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])
colors = np.array([0, 10, 20, 30, 40, 45, 50, 55, 60, 70, 80, 90, 100])

plt.scatter(x, y, c=colors, cmap='viridis',s=100)

plt.colorbar()

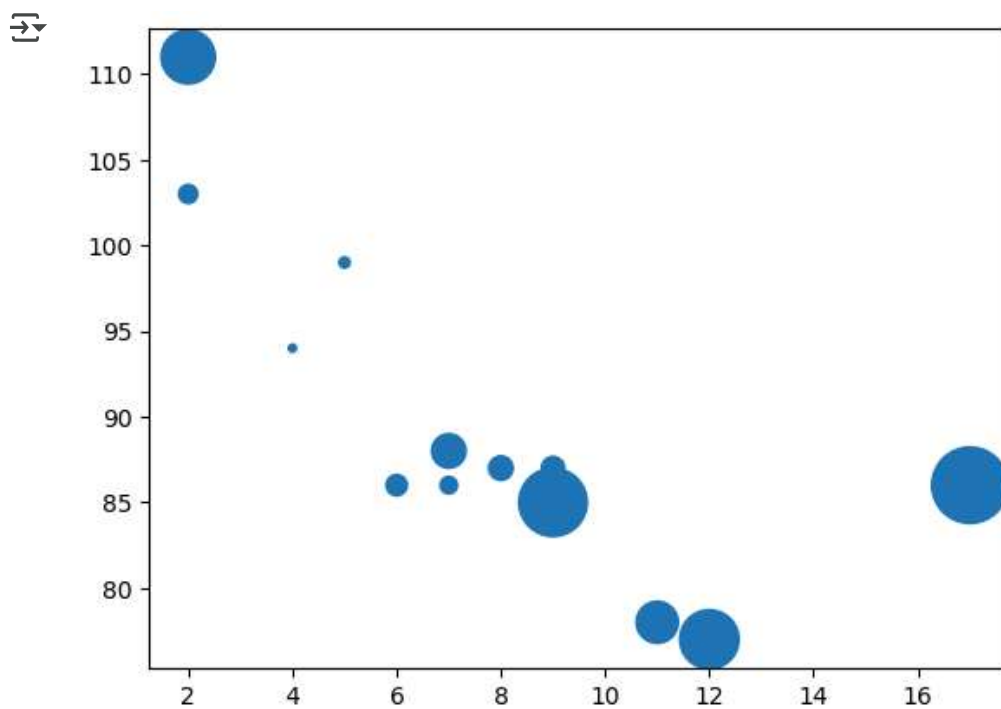
plt.show()
```



```
x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])
sizes = np.array([20,50,100,200,500,1000,60,90,10,300,600,800,75])
```

```
plt.scatter(x, y, s=sizes)
```

```
plt.show()
```

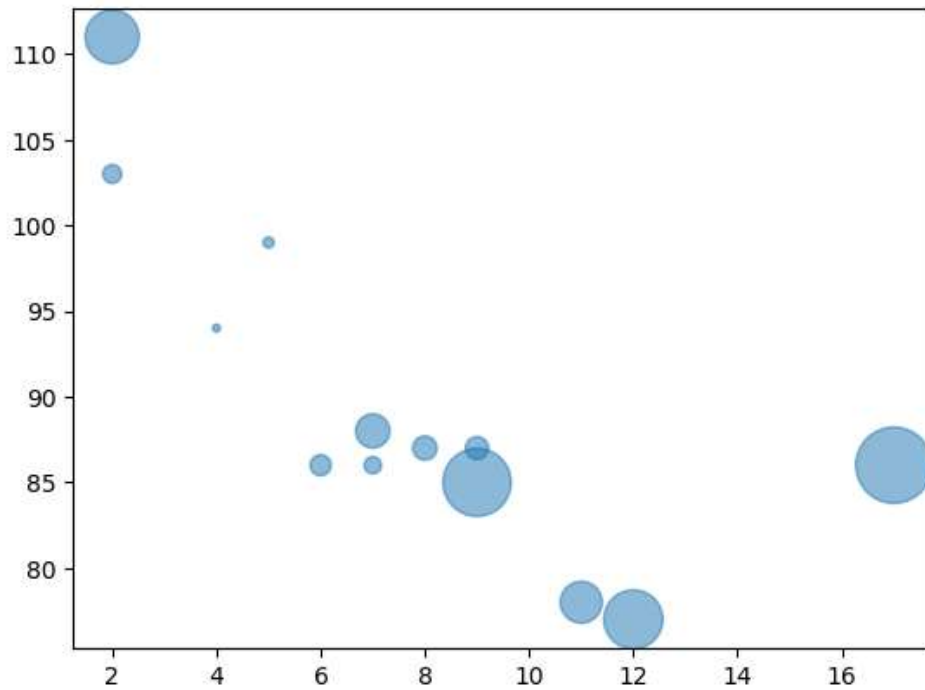


✓ Set transparency for dots

```
x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])
sizes = np.array([20,50,100,200,500,1000,60,90,10,300,600,800,75])
```

```
plt.scatter(x, y, s=sizes, alpha=0.5)
```

```
plt.show()
```

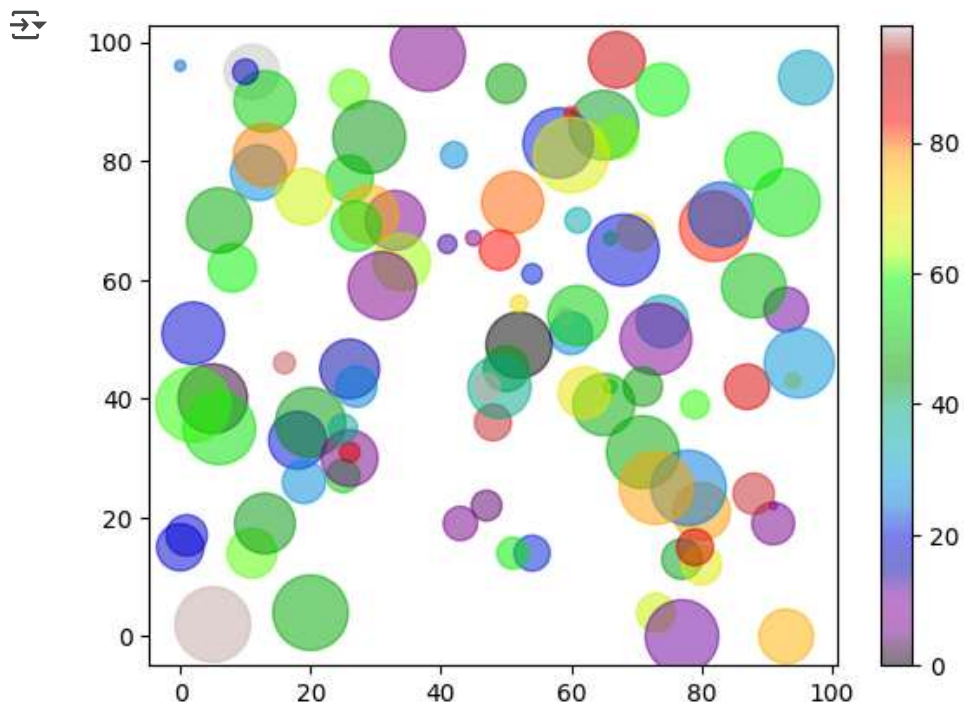


```
x = np.random.randint(100, size=(100))
y = np.random.randint(100, size=(100))
colors = np.random.randint(100, size=(100))
sizes = 10 * np.random.randint(100, size=(100))
```

```
plt.scatter(x, y, c=colors, s=sizes, alpha=0.5, cmap='nipy_spectral')
```

```
plt.colorbar()
```

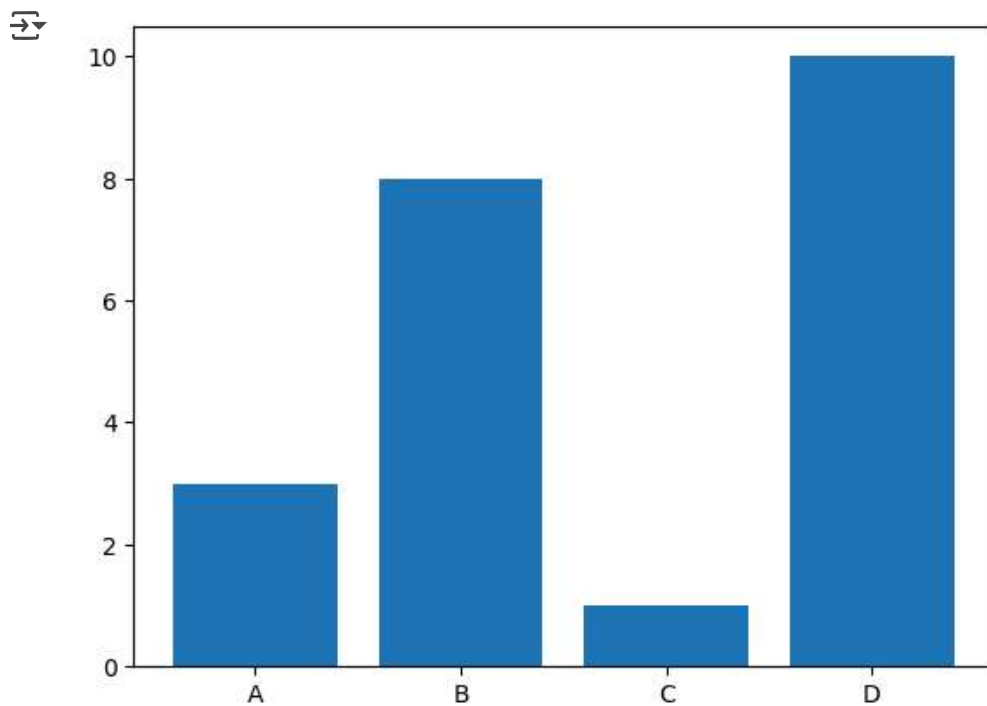
```
plt.show()
```



✓ Bar Plot

```
x = np.array(["A", "B", "C", "D"])
y = np.array([3, 8, 1, 10])
```

```
plt.bar(x,y)
plt.show()
```

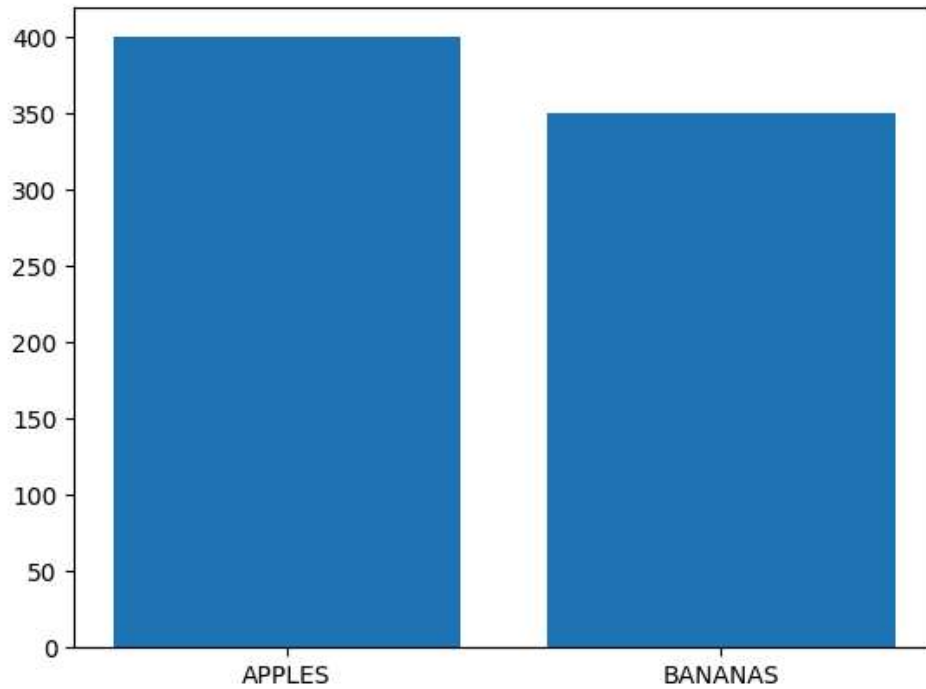


```
x = ["APPLES", "BANANAS"]
y = [400, 350]
```



```
plt.bar(x, y)
```

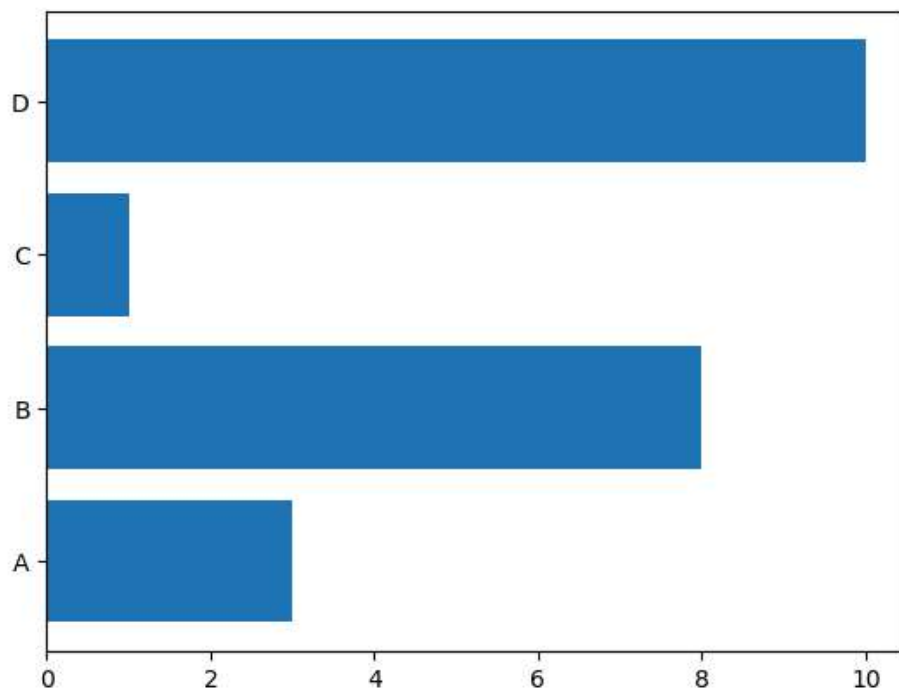
↔ <BarContainer object of 2 artists>



```
x = np.array(["A", "B", "C", "D"])
y = np.array([3, 8, 1, 10])
```

```
plt.barh(x, y)
plt.show()
```

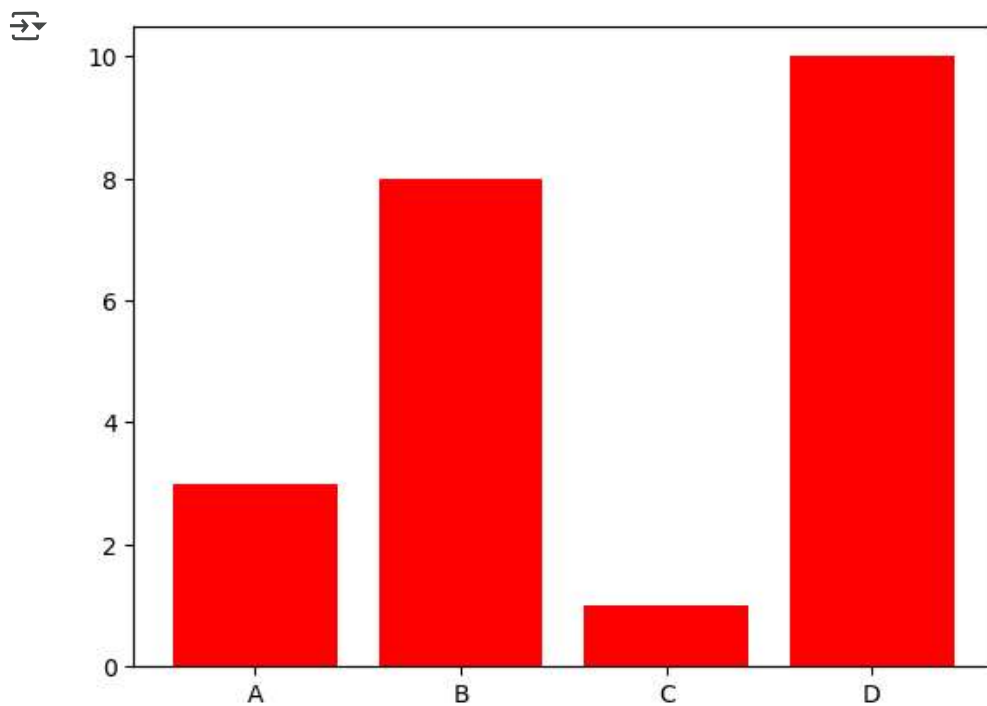
↔



✓ Bar color

```
x = np.array(["A", "B", "C", "D"])
y = np.array([3, 8, 1, 10])
```

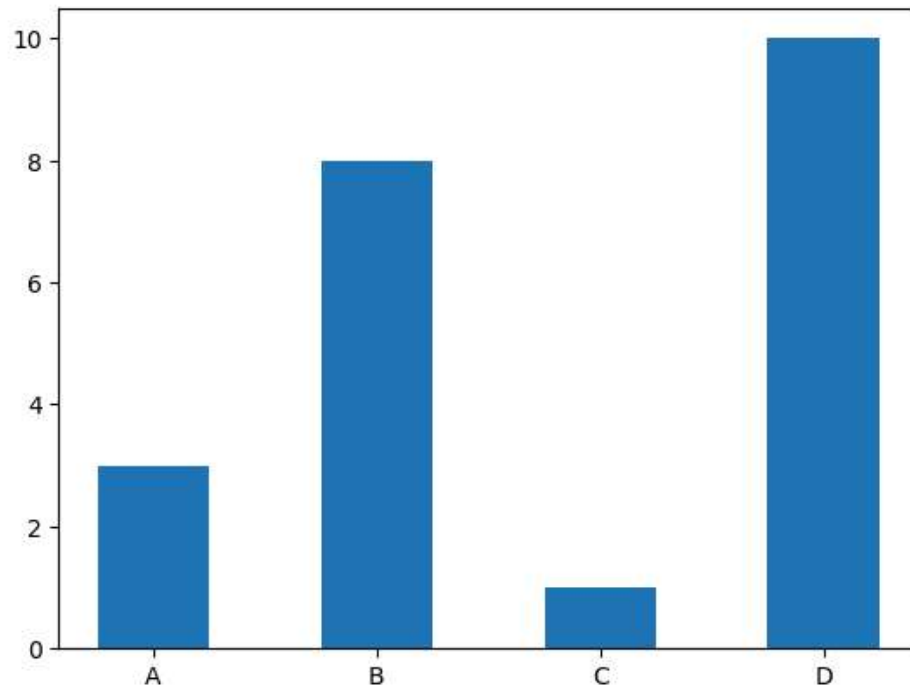
```
plt.bar(x, y, color = "red")
plt.show()
```



✓ Bar width

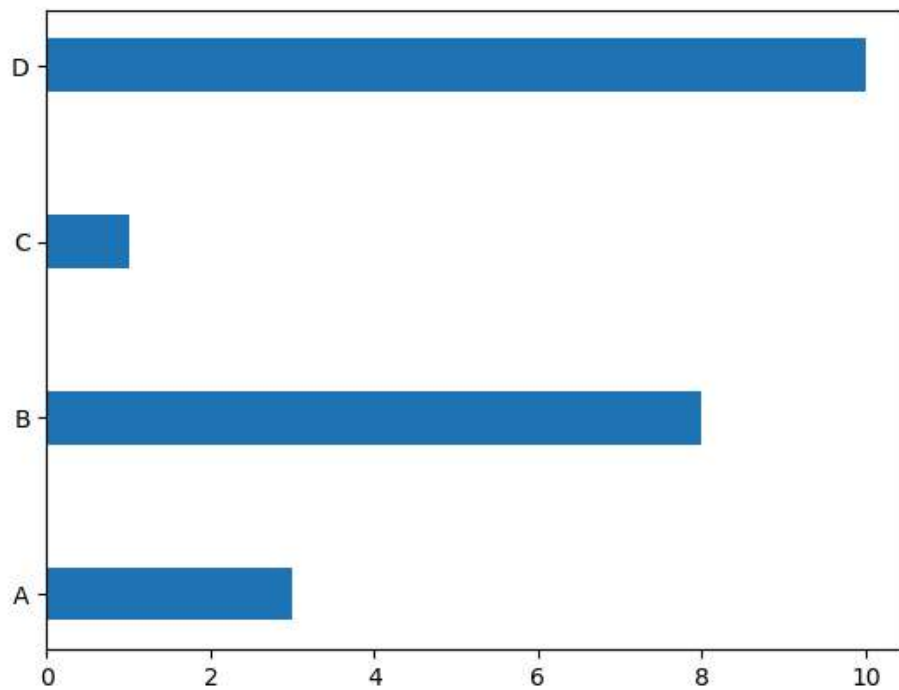
```
x = np.array(["A", "B", "C", "D"])
y = np.array([3, 8, 1, 10])
```

```
plt.bar(x, y, width = 0.5)
plt.show()
```



```
x = np.array(["A", "B", "C", "D"])
y = np.array([3, 8, 1, 10])
```

```
plt.barh(x, y, height = 0.3)
plt.show()
```



✓ Histogram plot

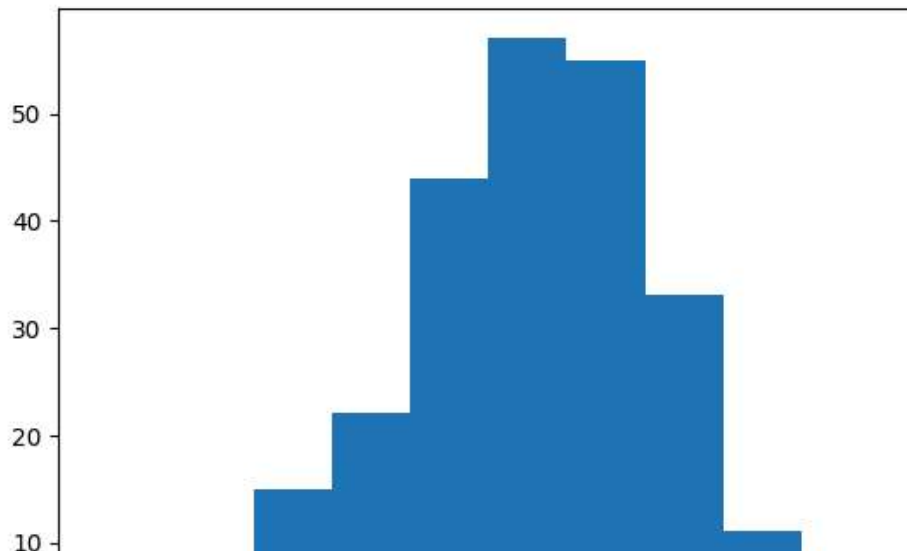
```
x = np.random.normal(170, 10, 250)
```

```
print(x)
```

```
→ [153.12439896 155.80478407 145.40444041 162.96441574 151.37457018  
164.15440483 183.3202973 177.26786621 157.39538364 184.71968724  
169.35587449 159.74364352 170.91835018 152.86175829 179.38818279  
166.37021046 163.08879968 155.86925151 162.63650422 174.47568621  
157.73287617 172.66655769 181.02672259 187.55718395 178.78356612  
166.21933544 162.92919088 150.40240305 159.45776114 176.35405853  
181.67876545 169.65059378 164.77413474 175.3958327 166.95403644  
166.95522827 176.87409107 160.49218243 185.0294758 178.67868416  
177.46010716 163.75814589 173.13363008 163.51880022 154.9474543  
169.57283706 162.80374596 163.0888315 170.67115826 180.28065837  
174.27264853 154.90726651 165.66699358 171.43267044 162.28499087  
159.376216 180.79973989 174.94686411 192.74710559 167.1172077  
179.69402204 171.94987252 166.27651995 175.75551048 172.25771258  
168.12355557 168.57645151 182.06829439 173.84508958 160.20599726  
156.59836432 163.79933511 164.8785034 156.97071535 160.42614925  
166.68037475 170.72134964 180.31761526 165.45527278 181.57436945  
169.77138255 173.10833195 164.75988804 158.73582206 174.89422607  
161.24572058 186.17164638 169.98469563 177.9056987 171.29897316  
171.4186769 160.05748997 174.82679163 175.95623137 168.25222583  
157.37983737 183.22923742 177.23494348 182.33116606 170.572398  
177.29256136 167.49185954 166.70320505 156.7734282 164.01514052  
167.87612062 161.62263602 172.20120686 176.69431928 173.02802608  
178.02941071 172.15722728 137.77706167 185.09011123 173.95296546  
148.33789895 182.49142019 175.82283639 177.13211667 177.42394622  
188.3411666 179.64525855 166.34191409 169.88697234 182.99076571  
171.22747419 175.46187016 151.0167871 184.54717871 165.23280617  
182.61265762 151.47982238 162.21732746 179.20037345 161.41221904  
162.92460715 185.37114288 160.96493745 176.44358058 147.89043937  
172.04096361 169.90318518 186.56343094 157.94013406 182.00113966  
168.10618904 188.23548251 155.35704714 173.49135184 160.52138556  
166.02614761 163.26638449 174.74226497 179.10188188 186.17404893  
169.21052955 156.64970072 161.40786235 181.61714019 179.08445525  
162.24612005 165.08904693 167.06640432 146.0330082 161.21191466  
168.27713872 154.93343166 172.49512558 168.60701155 177.13880459  
169.00591325 154.90081247 152.17611859 151.66452506 176.43669616  
161.19644045 181.35201308 173.47336022 165.70536759 164.84246539  
159.22545533 175.11847254 149.89693108 161.67294706 167.49553371  
162.47281021 161.4365345 156.96863657 163.74567039 171.64256114  
169.02856164 156.57637187 165.312846 183.40929764 163.50819022  
171.01214415 160.28438645 167.61508511 173.34554016 177.4586629  
159.64443608 173.29035109 174.82976164 174.30166877 159.84152933  
170.40714991 161.02006705 165.35191385 163.44910137 169.7272019  
178.16770301 172.29749089 171.17386424 169.10548156 170.33443639  
189.72043713 158.80423599 170.32044123 174.88563167 159.89676958  
167.57882964 166.54059551 182.76588906 160.71478502 168.50514757  
178.7534922 168.07581566 154.20627187 170.66568658 187.24085413  
155.71225695 166.00932243 165.53075898 172.65517782 163.72114184  
183.4018708 158.20207302 165.52328495 160.0275102 160.66719267  
163.69031162 175.03151676 163.71227652 170.2944158 166.97310009  
158.30220332 175.7503952 160.88679309 171.89479268 146.05553788]
```

```
x = np.random.normal(170, 10, 250)
```

```
plt.hist(x)  
plt.show()
```



```
x = np.random.normal(170, 10, 250)
```

```
plt.hist(x, bins=20, color='skyblue', edgecolor='black')  
plt.title("Histogram of Normally Distributed Data")  
plt.xlabel("Value")  
plt.ylabel("Frequency")  
plt.show()
```

